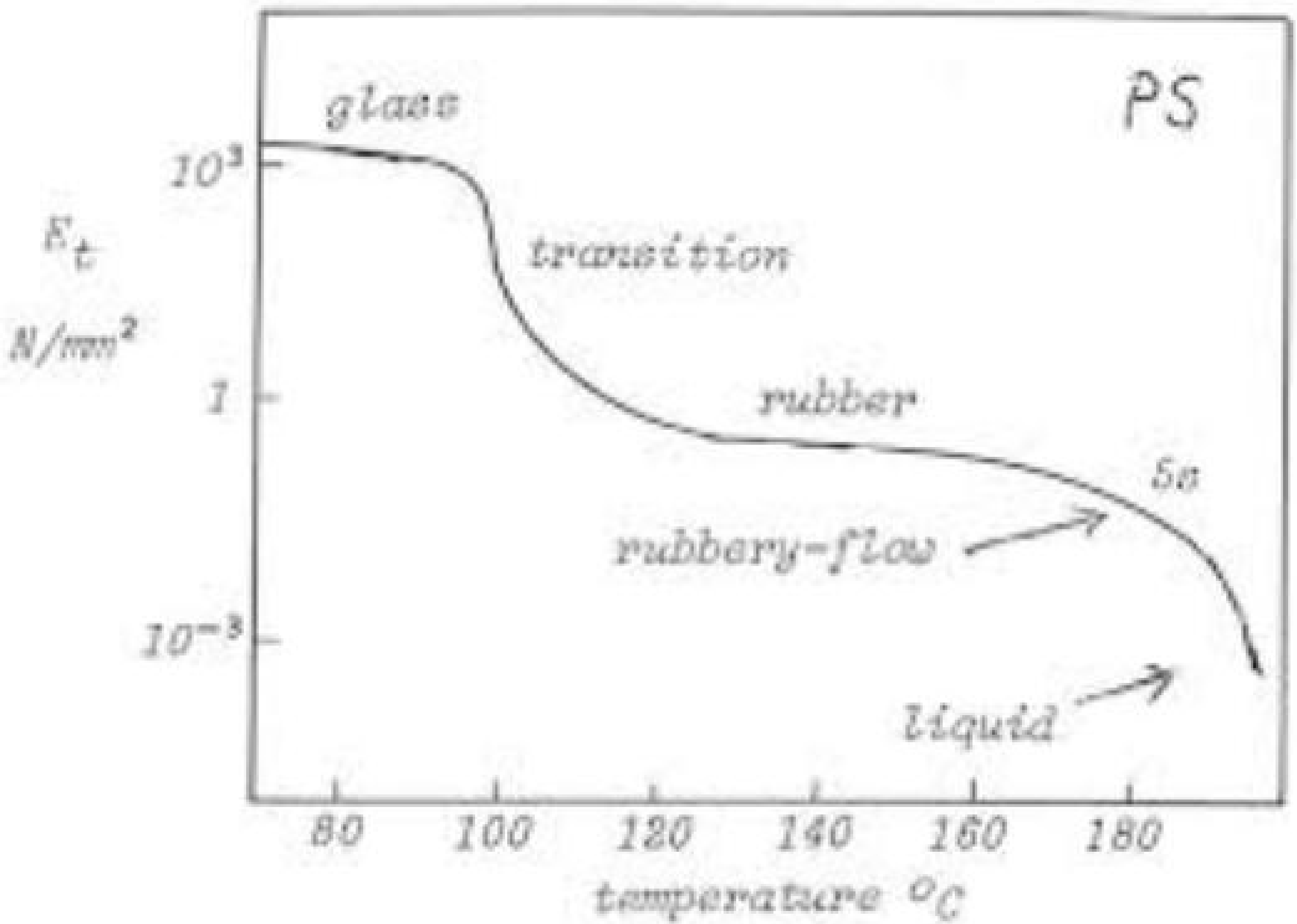
How to succeed

- Read the Chapter ahead of lectures
- Come to class
- **Start paper early**
- Study groups
- Prepare exams
- Don’t cheat
- **Learn with Internet: Wikipedia, google(baidu)…**
- Ask questions and Think skeptically

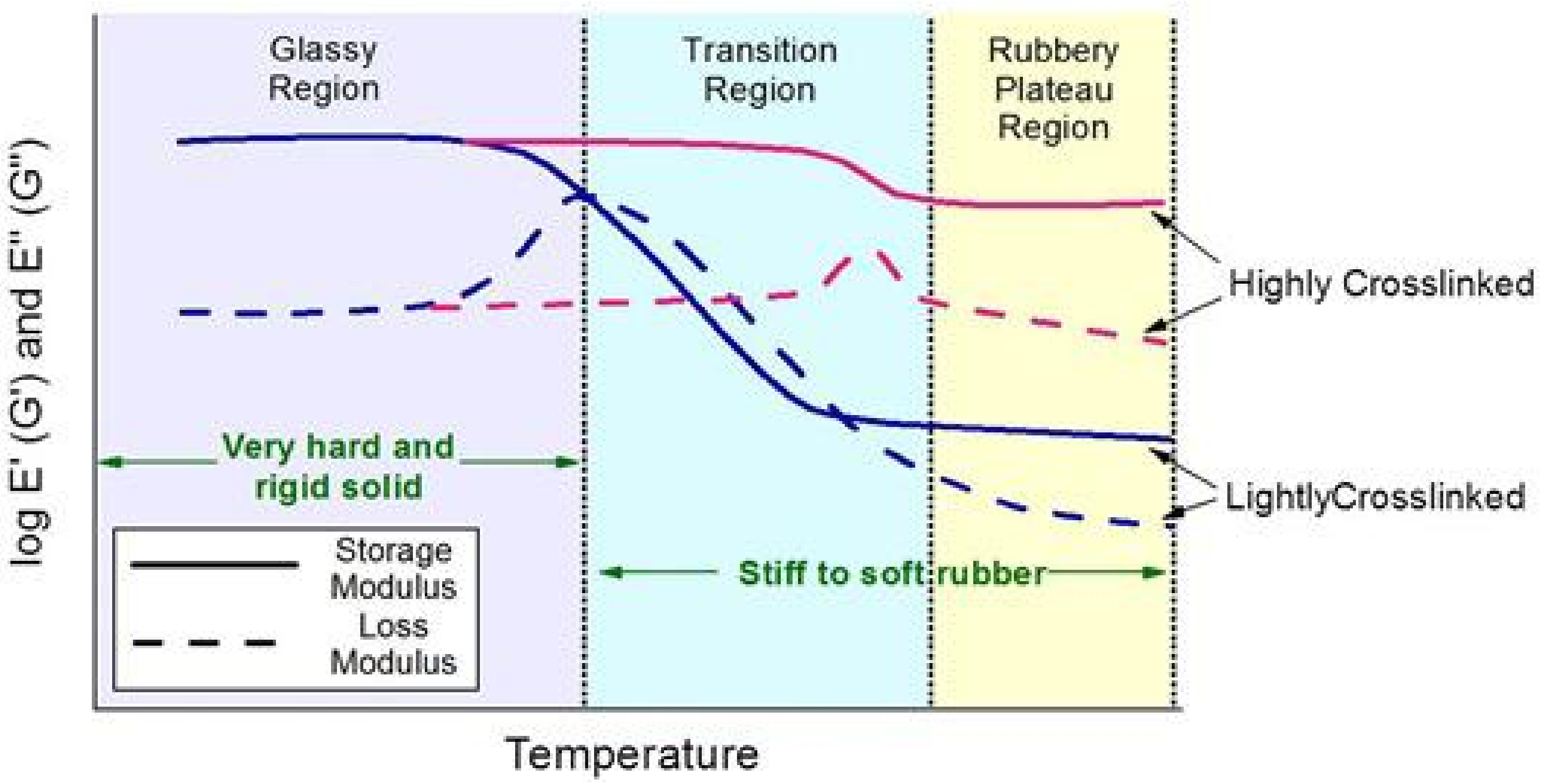
Lecture 13: Thermoplastics and Their Processing

SUN Dazhi

Thermoplastics



Thermosets



Glassy Region
Transition Region
Rubbery Plateau Region
Highly Crosslinked
LightlyCrosslinked
Very hard and rigid solid
Stiff to soft rubber
Storage Modulus
Loss Modulus
log E' (G') and E'' (G'')
Temperature

Thermoplastics

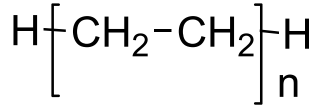
Polymers that are amorphous & glassy or semicrystalline at operating temperature

- Are melt processed then cooled into new forms
- Can be re-heated and processed

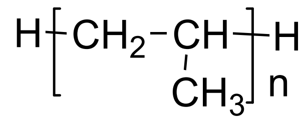
Reversible process

Commodity Thermoplastics

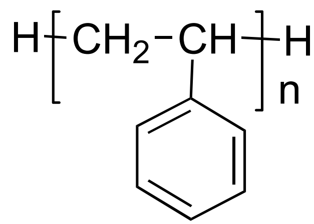
- Polyethylene



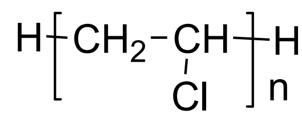
- Polypropylene



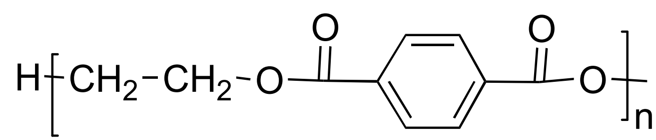
- Polystyrene



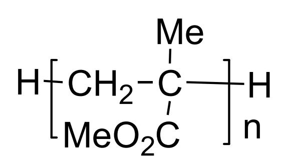
- Polyvinylchloride



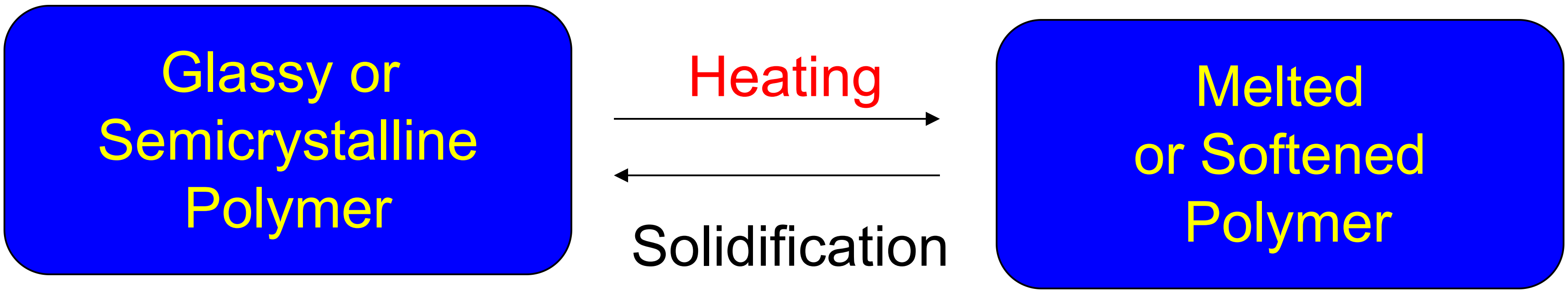
- Polyethylene terephthalate



- Polymethyl methacrylate

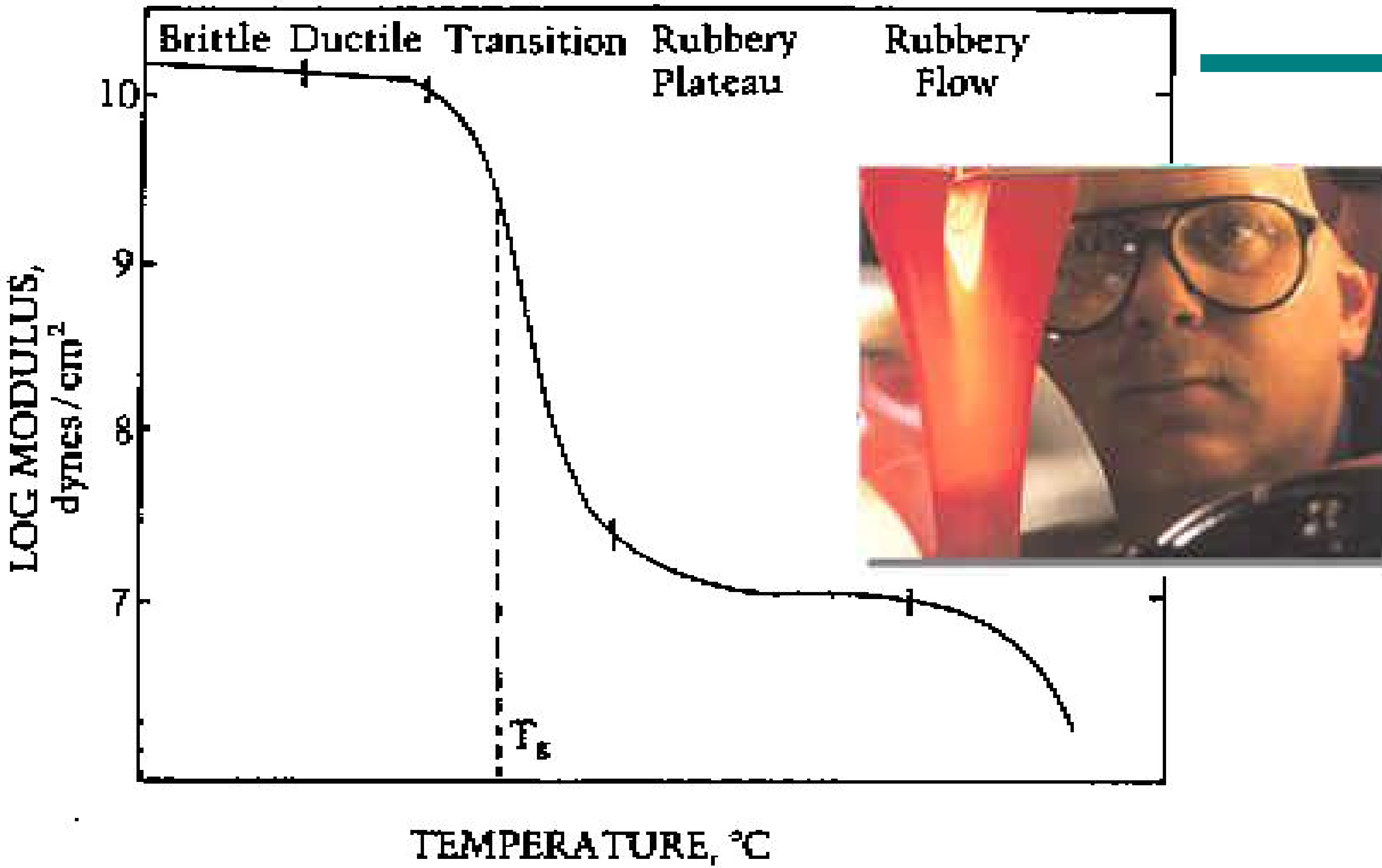


Thermoplastics

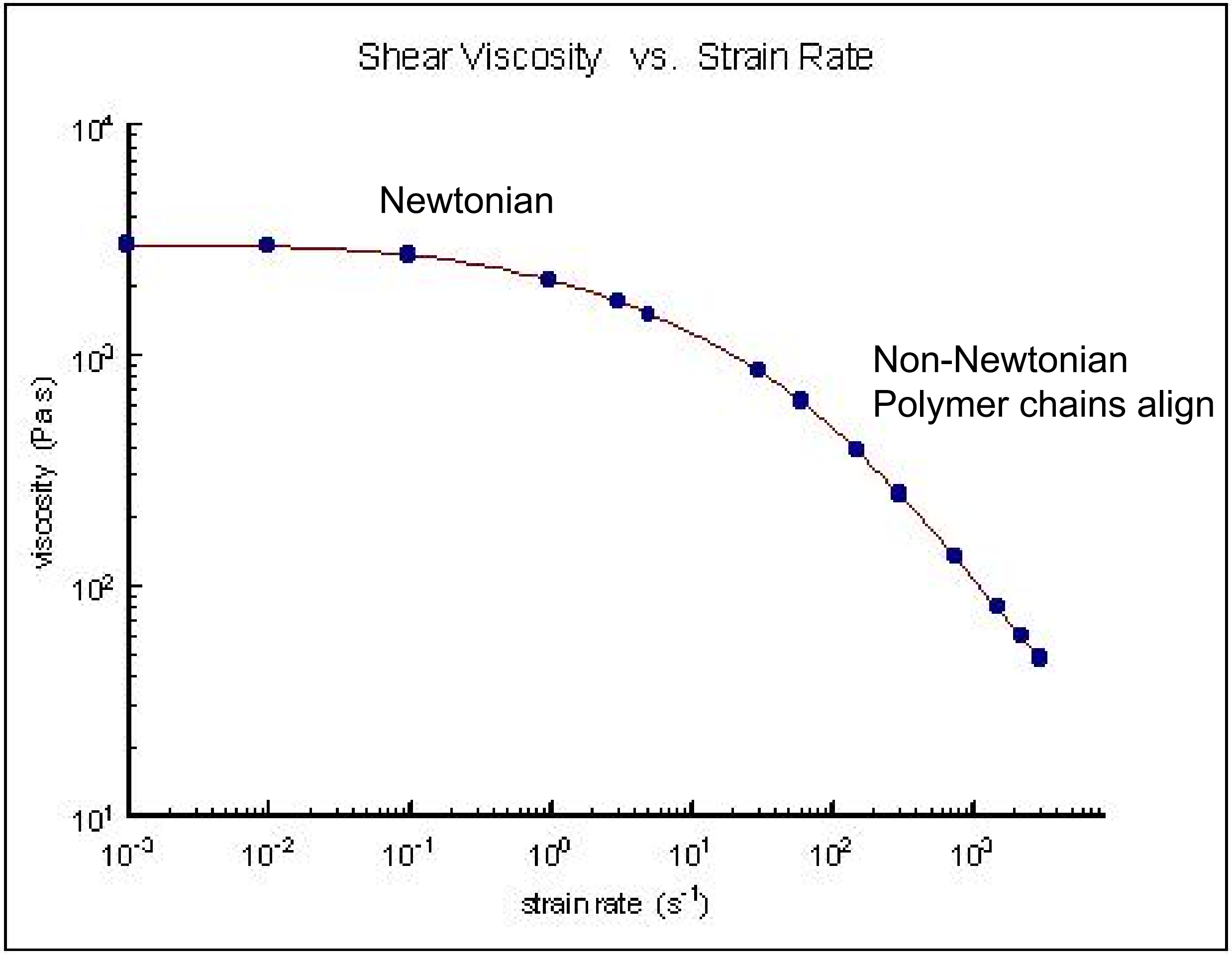


Extrusion through dies: linear products
Molding: Complex shapes

Linear vinyl polymers, Nylons, polyesters, polyarylenes polyimides.

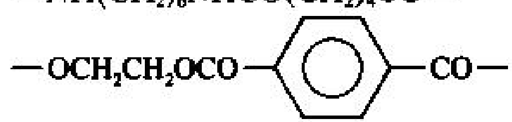
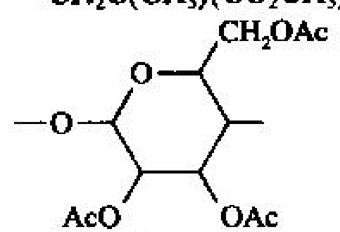

Brittle
Ductile
Transition
Rubbery Plateau
Rubbery Flow
10
9
8
7
T_g
LOG MODULUS, dynes/cm^2
TEMPERATURE, °C

Polymer Melt Viscosity decreases at High Shear

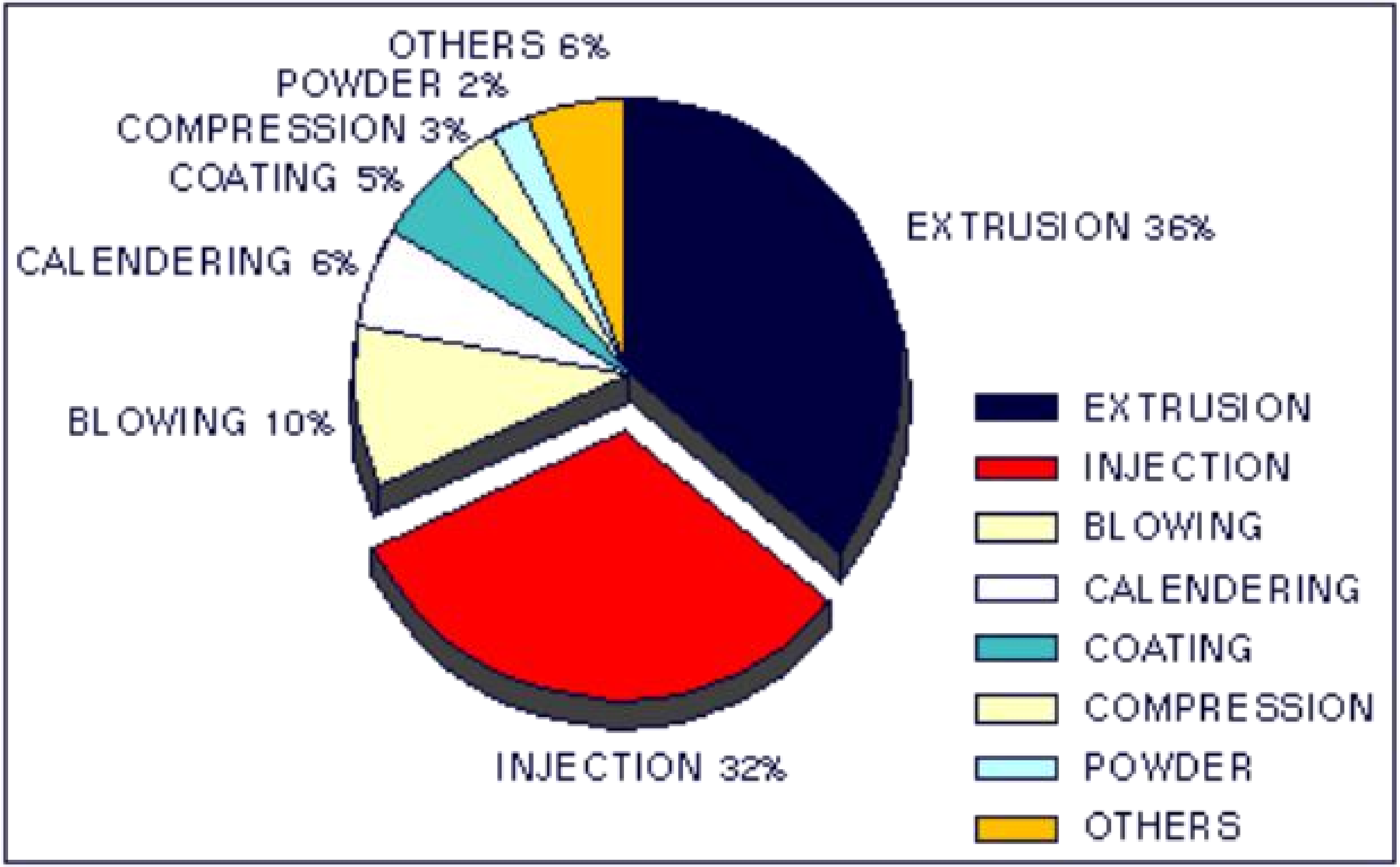


compression molding - 1-10 s^{-1}
calendering - 10-10^2 s^{-1}
extrusion - 10^2 -10^3 s^{-1}
injection molding - 10^3 -10^4 s^{-1}

100 and above generally out of Newtonian range

Polymer	Repeating Unit	T_g (°C)	T_m (°C)
Polydimethylsiloxane	$—OSi(CH_3)_2—$	−127	−40
Polyethylene	$—CH_2CH_2—$	−125	137
Polyoxymethylene	$—CH_2O—$	−82	181
Natural rubber (polyisoprene)	$—CH_2C(CH_3){=}CHCH_2—$	−73	28
Polyisobutylene	$—CH_2C(CH_3)_2—$	−73	44
Poly(ethylene oxide)	$—CH_2CH_2O—$	−41	66
Poly(vinylidene fluoride)	$—CH_2CF_2—$	−40	185
Polypropylene	$—CH_2CH(CH_3)—$	−8	176
Poly(vinyl fluoride)	$—CH_2CHF—$	41	200
Poly(vinylidene chloride)	$—CH_2CCl_2—$	−18	200
Poly(vinyl acetate)	$—CH_2CH(OCOCH_3)—$	32	
Poly(chlorotrifluoroethylene)	$—CF_2CFCl—$		220
Poly(ε-caprolactam)	$—(CH_2)_5CONH—$	52	223
Poly(hexamethylene adipamide)	$—NH(CH_2)_6NHCO(CH_2)_4CO—$	50	265
Poly(ethylene terephthalate)		61	270
Poly(vinyl chloride)	$—CH_2CHCl—$	81	273
Polystyrene	$—CH_2CHØ—$	100	250
Poly(methyl methacrylate)	$—CH_2C(CH_3)(CO_2CH_3)—$	105	200
Cellulose triacetate			306
Polytetrafluoroethylene	$—CF_2CF_2—$	117	327

Processes



OTHERS 6%
POWDER 2%
COMPRESSION 3%
COATING 5%
CALENDERING 6%
BLOWING 10%
EXTRUSION 36%
INJECTION 32%
EXTRUSION
INJECTION
BLOWING
CALENDERING
COATING
COMPRESSION
POWDER
OTHERS

EXTRUSION

Shaping through die

Final Product (pipe, profile)

Secondary operation

- **Fiber spinning (fibers)**
- **Cast film (overhead transparencies,**
- **Blown film (grocery bags)**

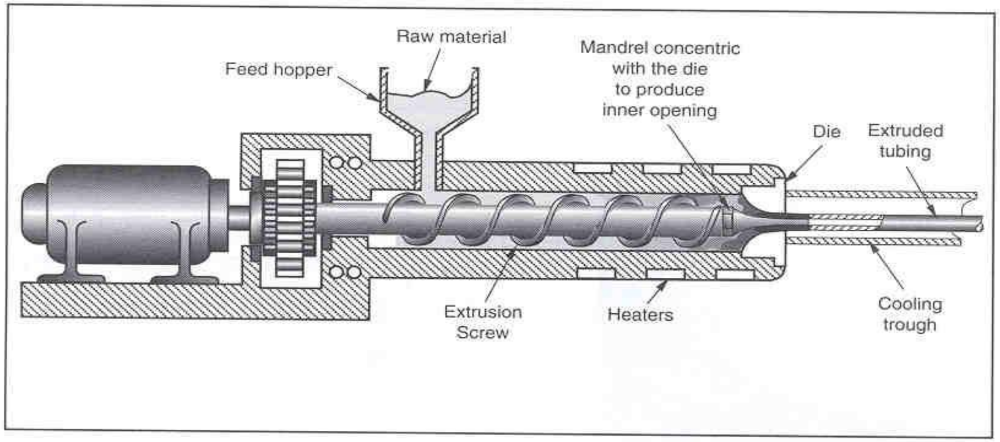
Preform for other molding processes

- **Blow molding (bottles),**
- **Thermoforming (appliance liners)**
- **Compression molding (seals)**

Extrusion

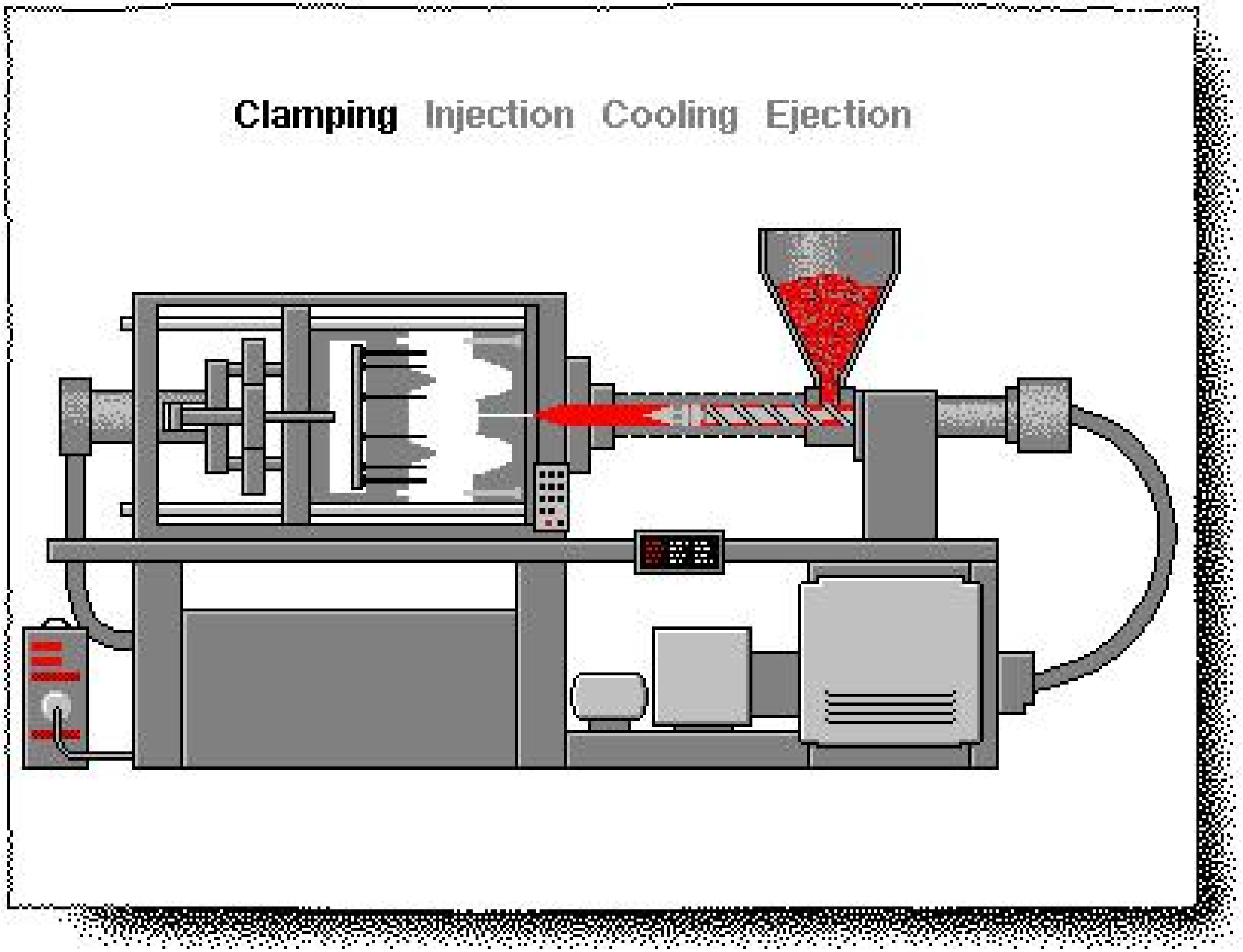
Extrusion process

- most common manufacturing of plastic resin
- combines pigments, additives and resin
- high heat, high pressure ➔ molten mixture
- pushed through die
- create warm plastic for possible further "finishing" operation such as pelletizing, calendaring, or molding

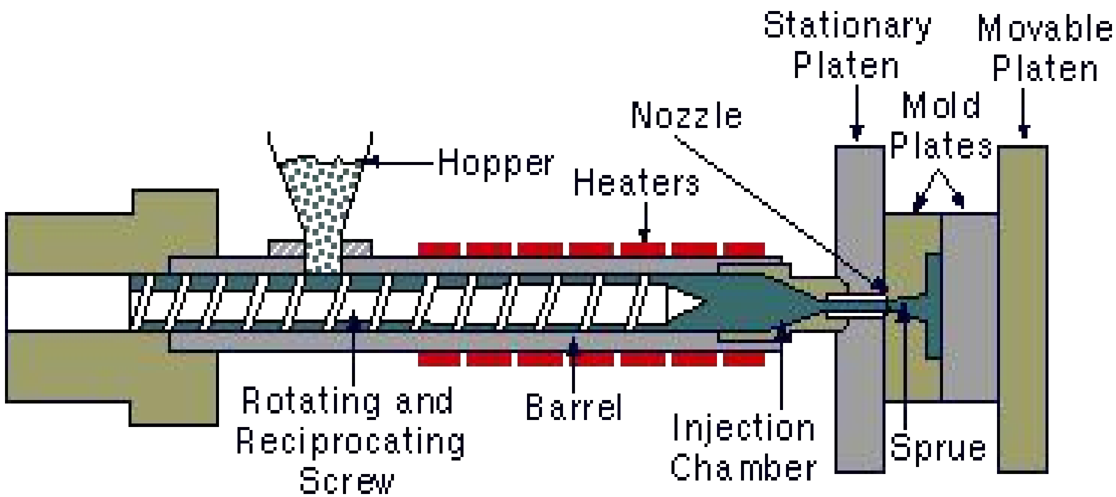


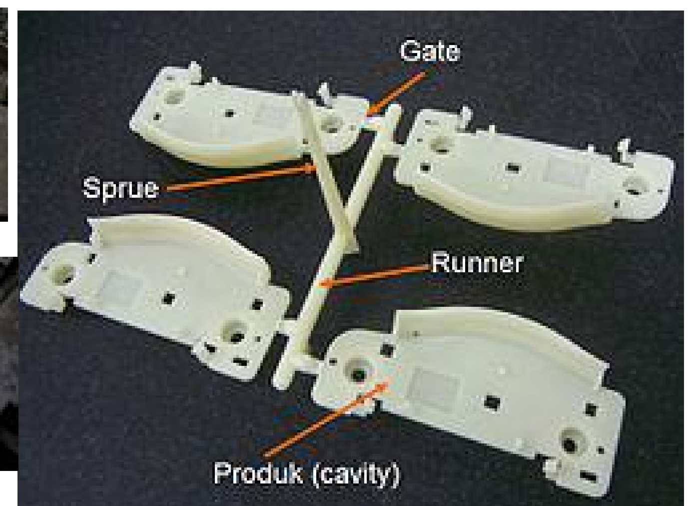
10-30 second cycle time

Injection Molding



Injection Molding Machine

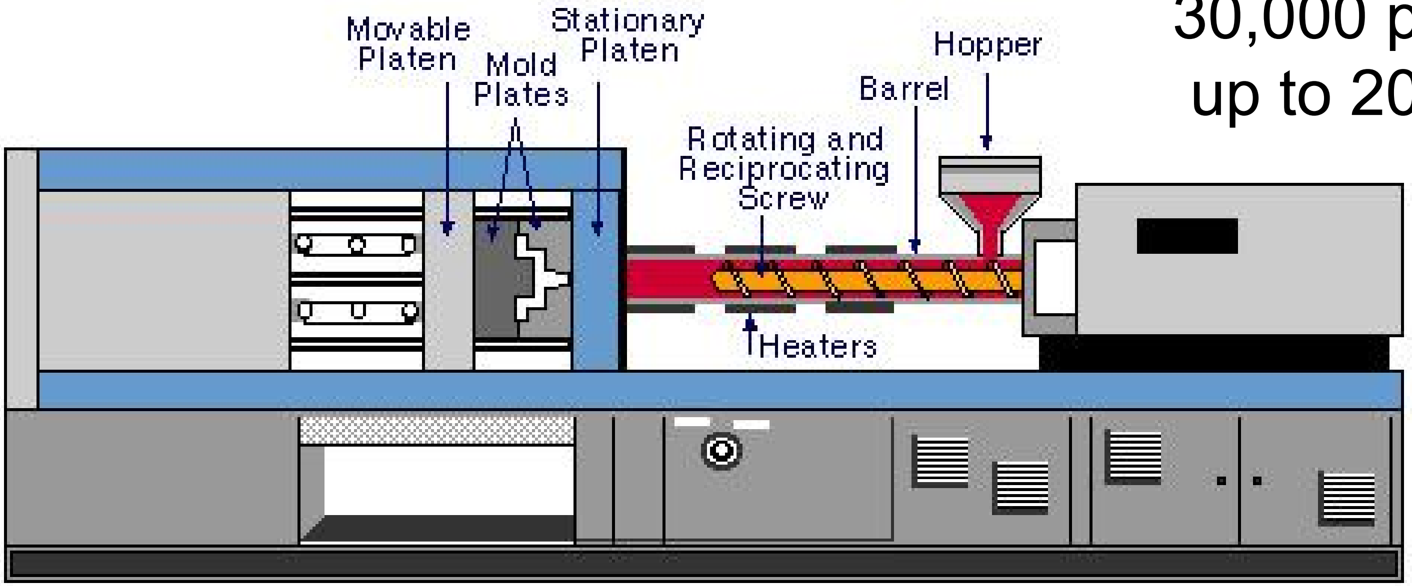




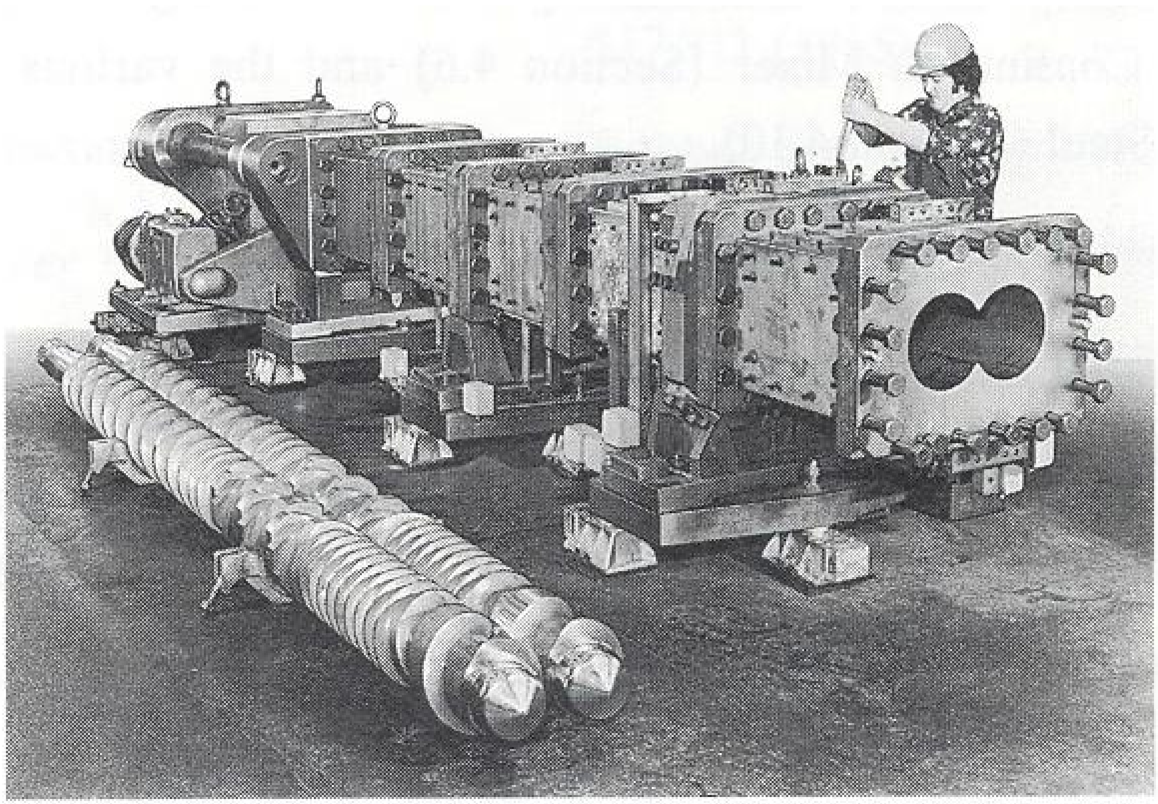
Injection Molding Machine

Injection Molding

30,000 psi
up to 200 MPa

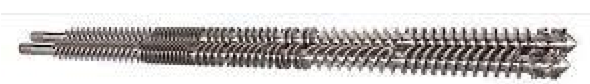


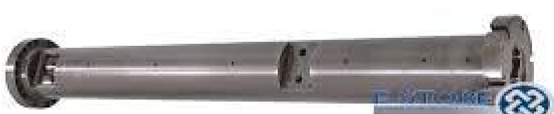
Parallel Twin Screw Extruder



Construction of ZSK 300 showing modular barrel and screws (courtesy of Werner and Pfleiderer).

Conical Twin Screw Extruder



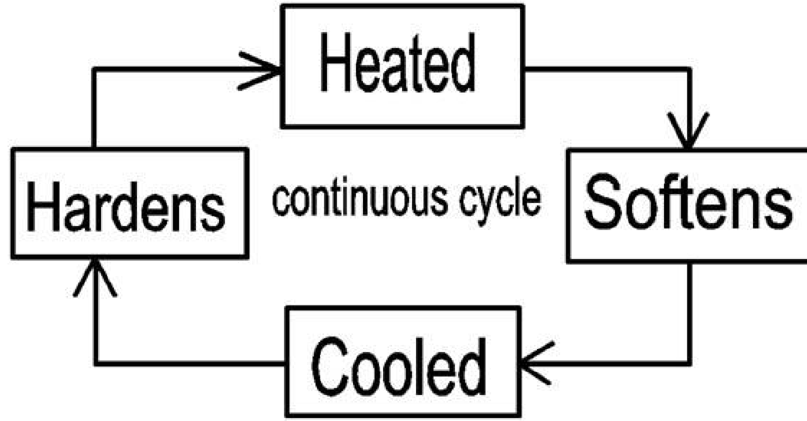


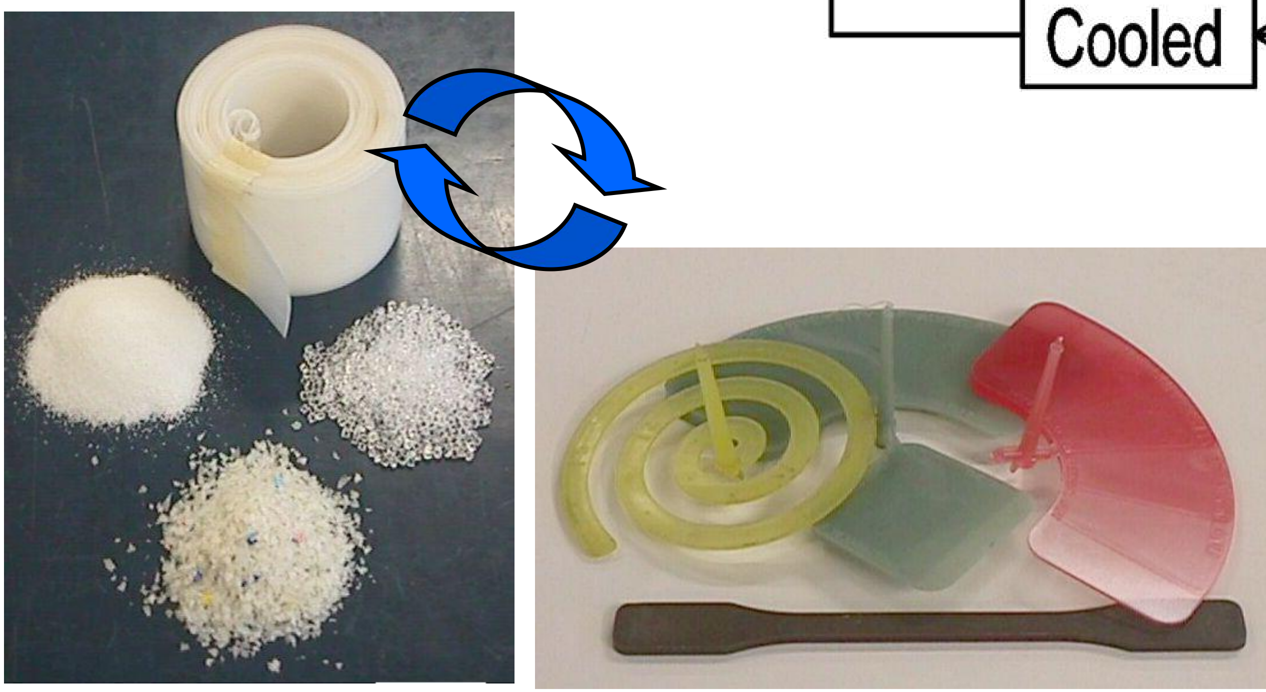
Processing Thermoplastics

Rule of Thumb

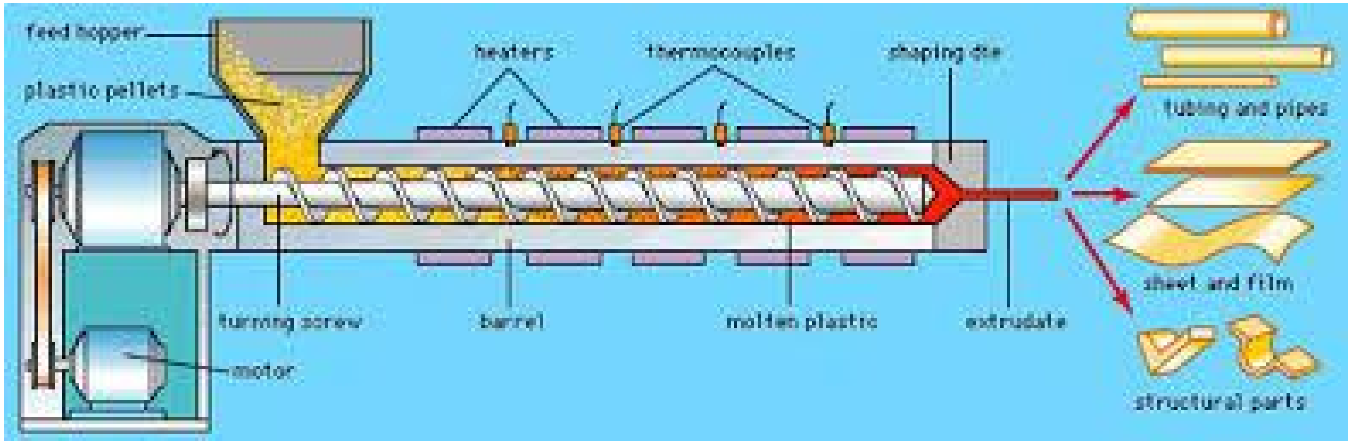
Amorphous: $T_g + 80\,°C$

Crystalline: $T_m + 30\,°C$

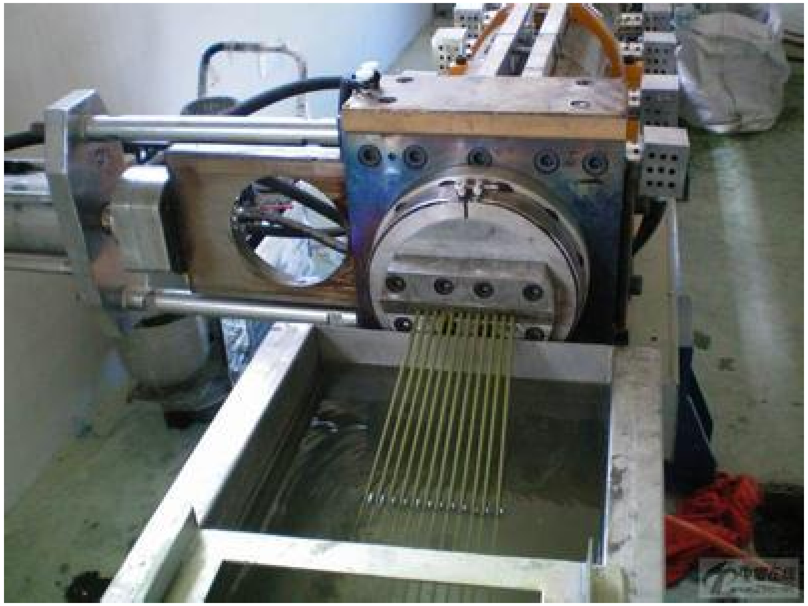




Pelletizing



Pelletizing



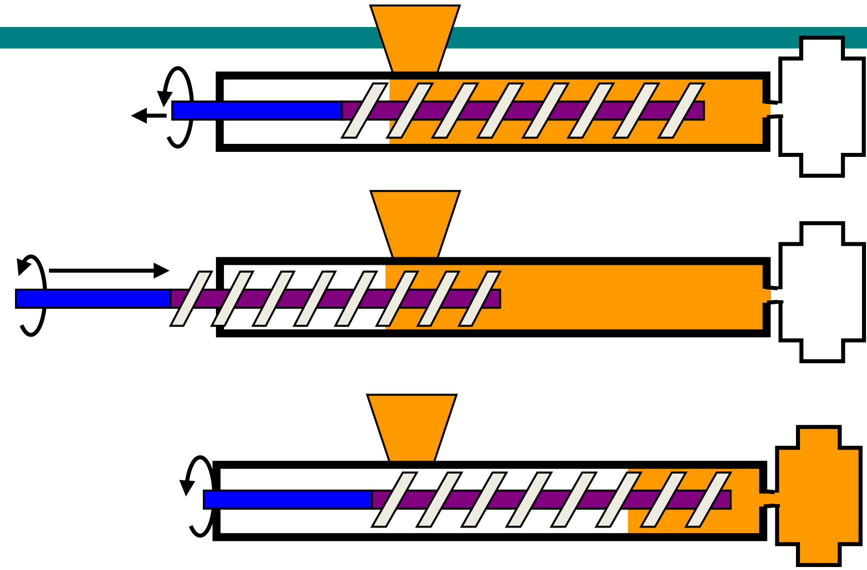
http://www.56.com/u40/v_ODE5OTg5NTc.html

http://v.ku6.com/show/HLXPAS_MT4XlKb8H.html open cross-sections (channels) | closed cross-sections (tubes, pipes) | pellets

http://baidu.ku6.com/watch/7738598549475884582.html?page=videoMultiNeed
http://v.ku6.com/show/JNMBEW-v5nbefCanoT4IMg...html

PRECISION TUBING LINE

Injection Molding of thermoplastics

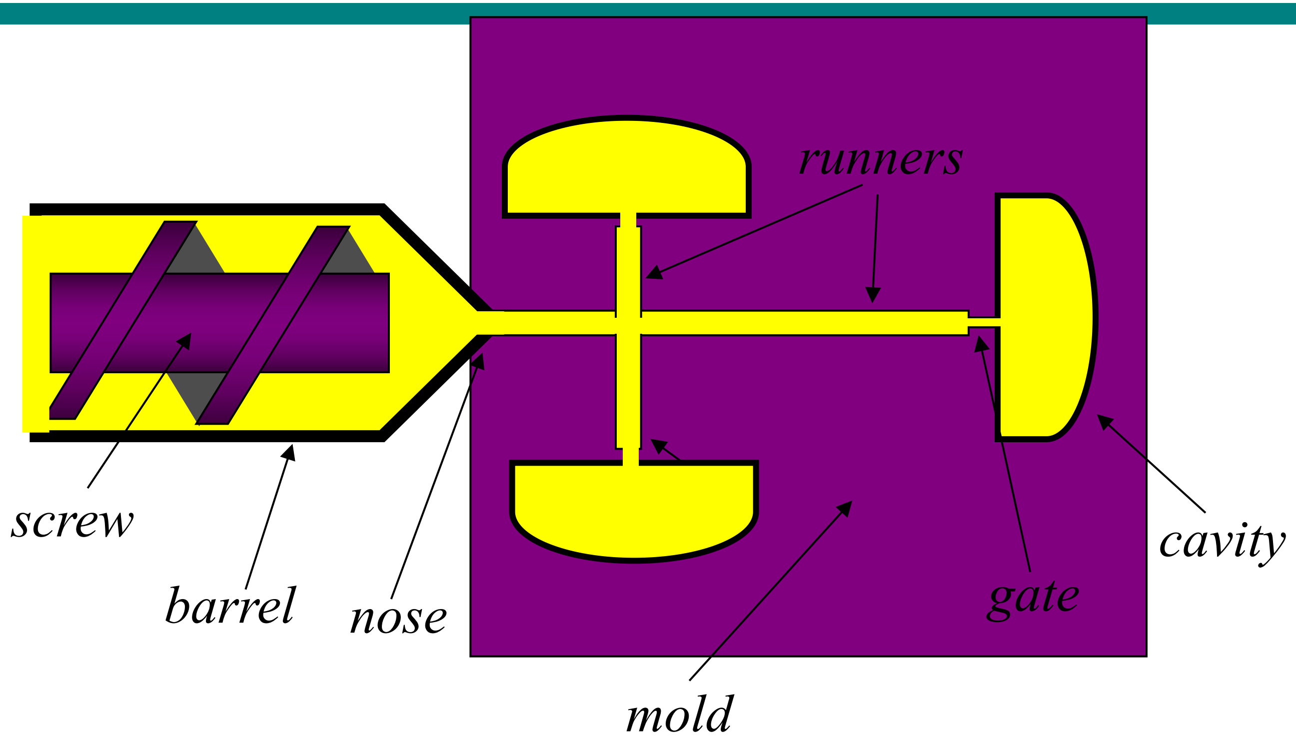


Rule of thumb: A pressure increase of 100 MPa, is equivalent to a temperature decrease of 50 ° C.

Shear thinning

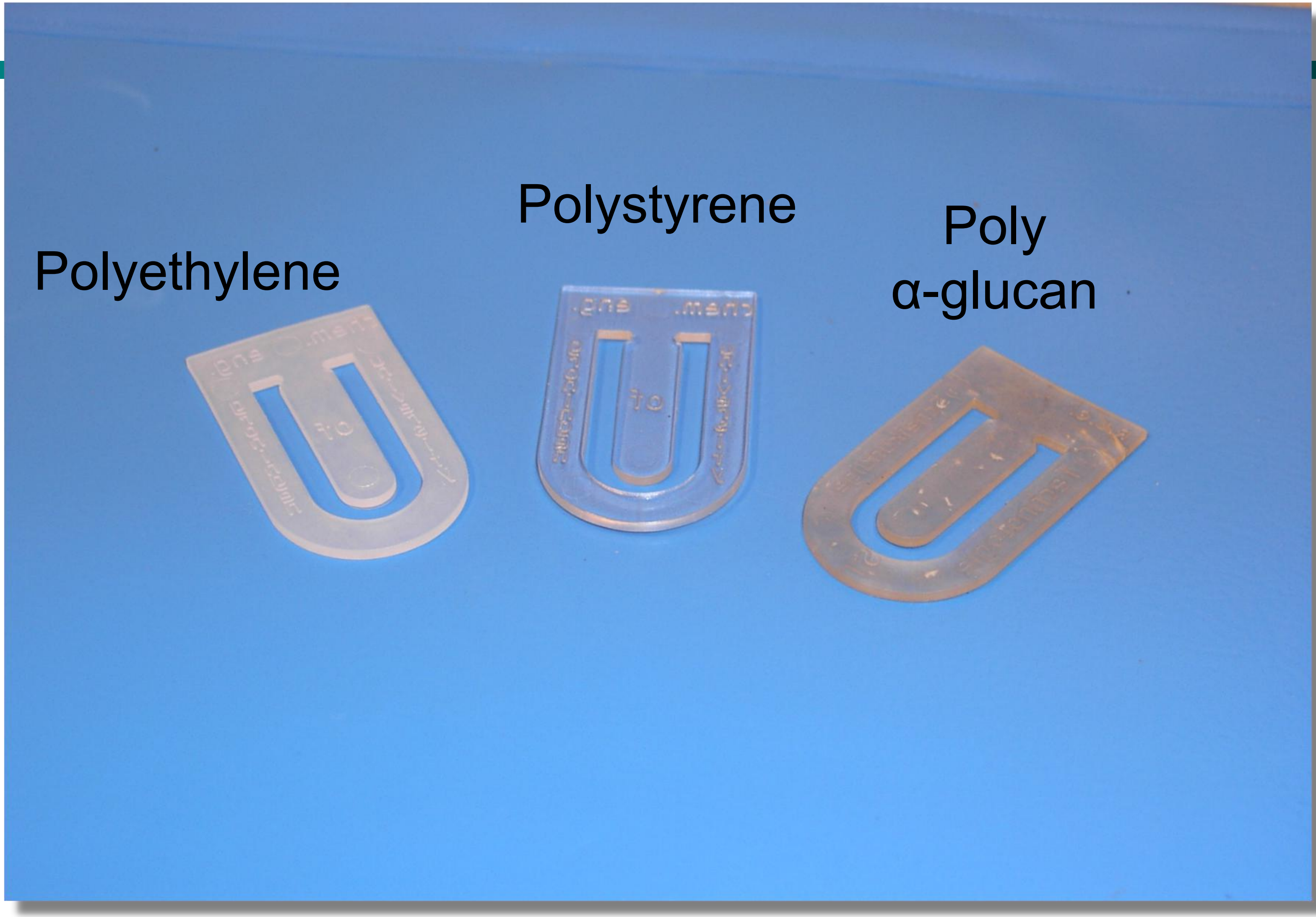
	Thermoplastic Injection Molding
Temperatures	
Materials	200-300 ^{0}C
mold	50-150 ^{0}C
Injection pressure	100 MPa
Viscosity	1000 Pa.s

Therminology



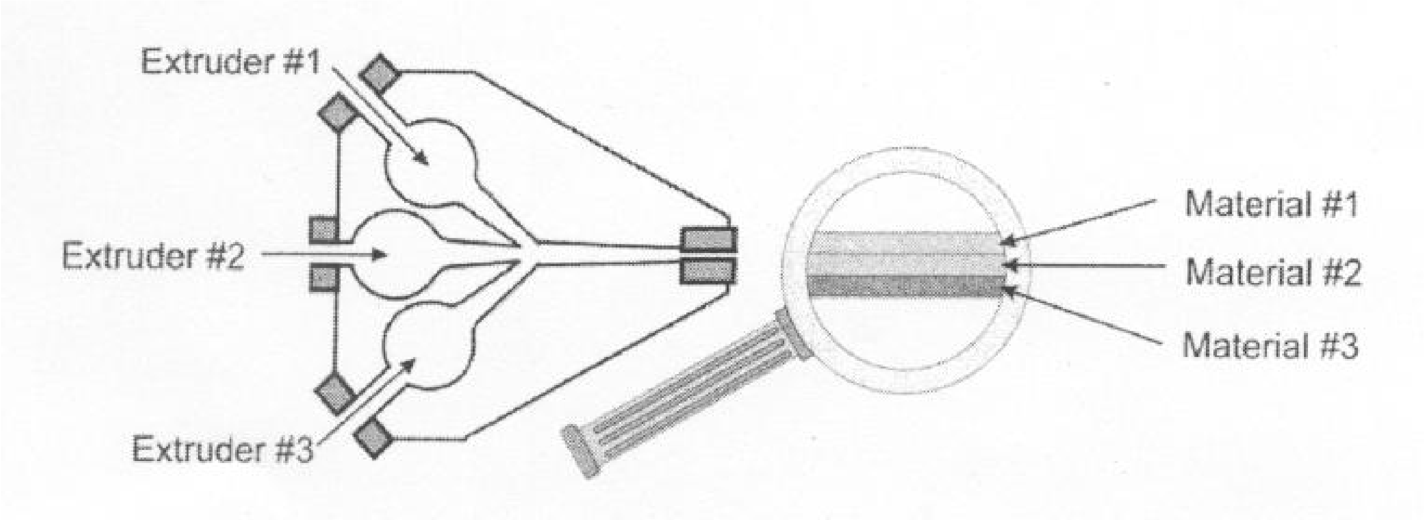
<u>Thermal effects:</u>

solidification during filling
viscous dissipation in the runners
shrinking during solidification
relaxation of the polymer
temperature fluctuations


Polyethylene
Polystyrene
Poly
α-glucan

Co-extrusion

- ***Co-extrusion***
  - **Two or more polymers extruded together through the same die**



http://v.ku6.com/show/tfkUG3SYvyUkx_yL7HkPqA...html

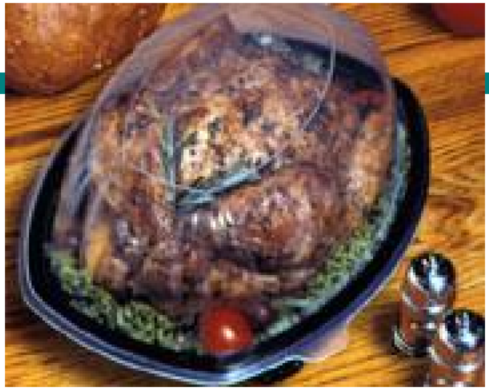
TABLE 4.2. *Common Multilayer Flexible Structures and Applications.*

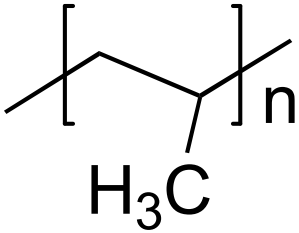
Multilayer Structure	Application
Two-side PVDC-coated Cellophane/PE/Foil/PE	Nut, dehydrated soup mixes dry chemicals, pharmaceutical powders
Glassine/LDPE/Foil/LDPE	Tobacco
Paper/Foil	Wrappers for chewing gum, cigarettes margarine/butter soap, carton overwrap, carton liner for sugar-coated cereal
Paper/LDPE/Foil/LDPE	Dry soup, drink mix
LDPE/LDPE	Rice
Nylon/PVDC/PE	Meat, cheese
Metallized Nylon/LDPE	Coffee
PVDC-coated Nylon/ Ionomer	Lunch meat and hot dogs
Polyester/PVDC/PE	Meat, cheese
PVDC-coated PET/LDPE	Processed meat
PVDC/PVC	Blister packaging
BOPP/LDPE	Candies, cookies
BOPP/PVDC-coated BOPP	Candy bar wraps

Handbook of Package Engineering-Hanlon, Kelsey & Forcino

Polypropylene

semicrystalline



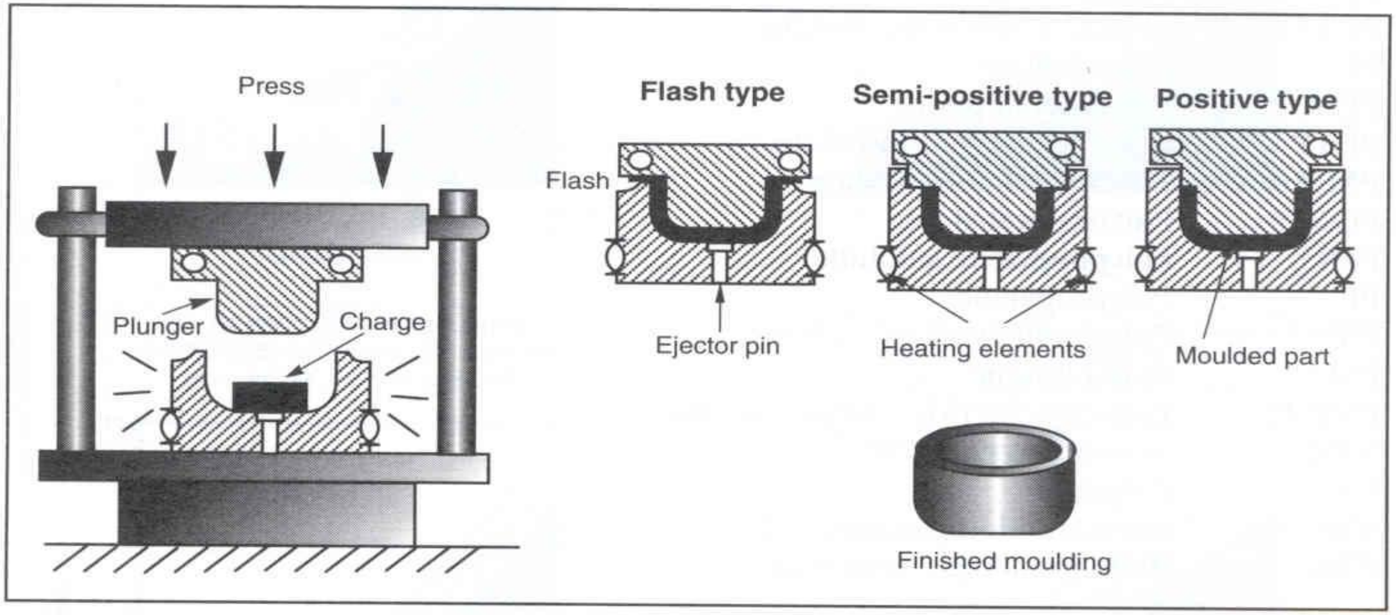


- Tensile Strength: 31-41 MPa, Modulus = 1.2-1.7 Gpa
- Elongation 100-600%

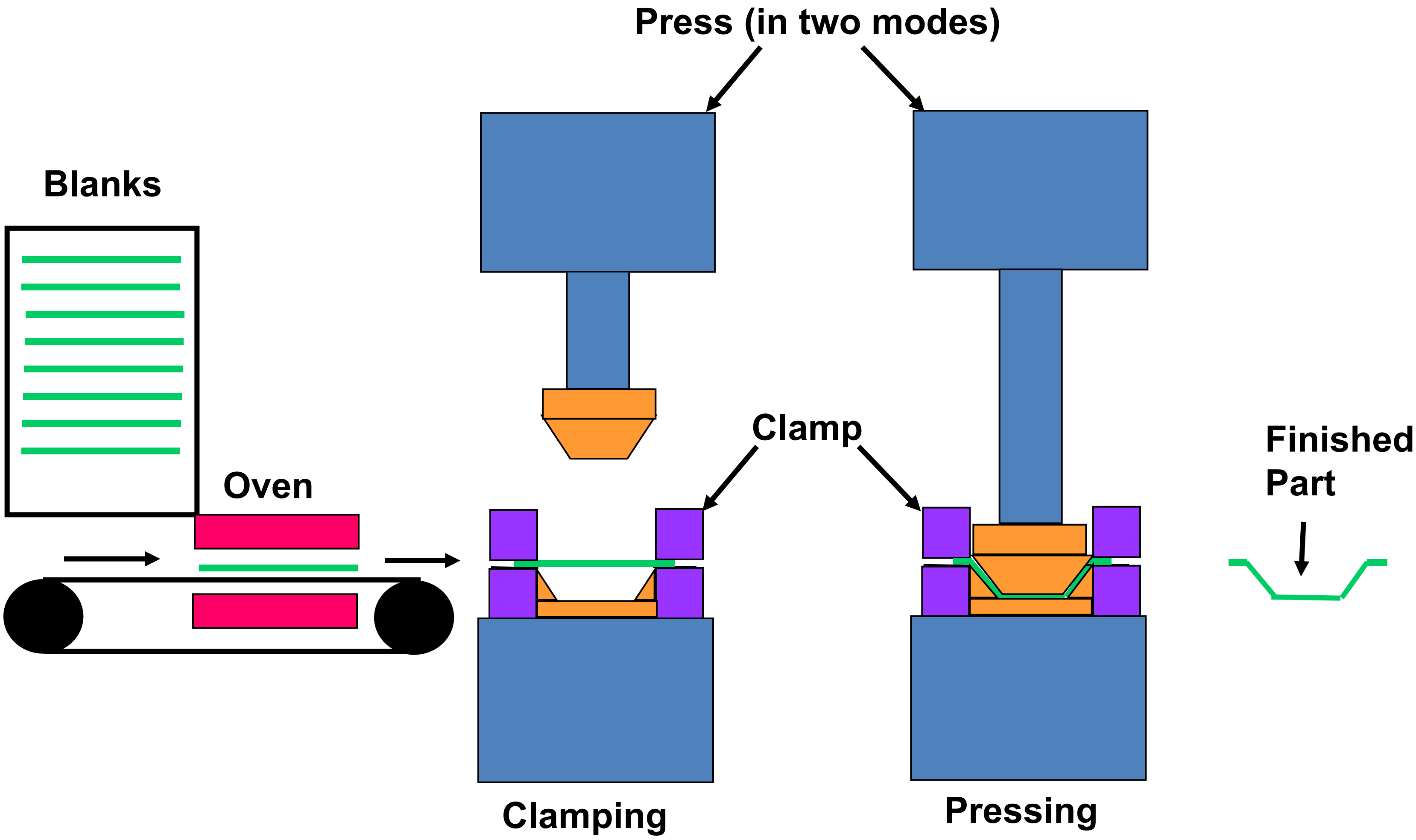
Living Hinge

Compression Molding

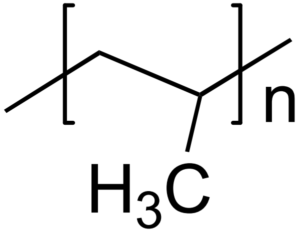
- Pre-shaped part placed directly into heated mold.
- Examples: Dishes, Handles, Electrical Components, Fittings and housings



Thermoplastic – Thermoforming

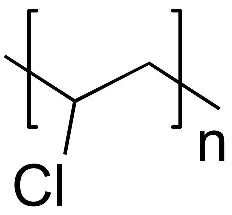


http://www.tudou.com/programs/view/c1dTMMQQ0ns



Thermoformed from sheet polypropylene

Polyvinyl Chloride



amorphous

- **No Plasticizer: Rigid Polymer (pipe)**

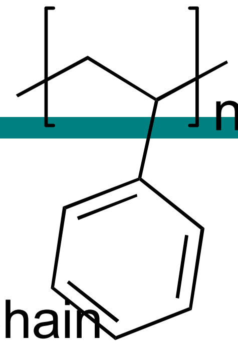
Tensile Strength: 65 MPa, Modulus = 3.5 GPa
Elongation 10%
Saran Wrap, floor tiles, bottles

- **40 wt% Plasticizer: soft pliable (Tygon tubing)**

Tensile Strength: 15 MPa
Elongation 400%
Synthetic leather, shower curtains

Polystyrene

Amorphous



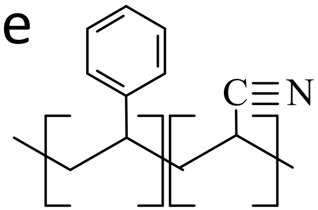
Commercial poly(styrene), PS, is a substantially linear, atactic polymer. Chain stiffness induced by the phenyl substituent creates a high Tg (105° C),

Tensile Strength: 45 MPa,
Modulus = 3.2 GPa
Elongation 4%

Styrofoam, molded objects such as tableware (forks, knives and spoons), trays, videocassette cases. Styrofoam, molded objects such as tableware (forks, knives and spoons), trays, videocassette cases.

ABS

–High strength, dimensional stability, impact resistance
–Poor UV resistance
–Telephones, PC housing & keyboards, ...



Grafted with polybutadiene

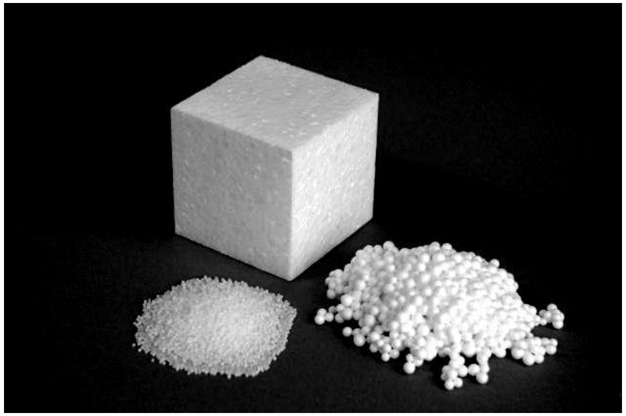
Property	ASTM	GP-PS	HIPS	ABS
Specific gravity	D792	1.04–1.05	1.03–1.06	1.03–1.58
Tensile strength, MPa[a]	D638	36.6–54.5	22.1–33.8	41.4–51.7
Tensile modulus, GPA[b]	D638	2.41–3.38	1.79–3.24	2.07–2.76
Elongation-to-break, %	D638	1–2	13–50	5–25
Impact strength, notched Izod, J m^{-1} [c]	D256	13.3–21.4	26.7–587	160–320
Heat-deflection temperature, °C at 455 kPa (66 psi)	D648	75–100	75–95	102–107

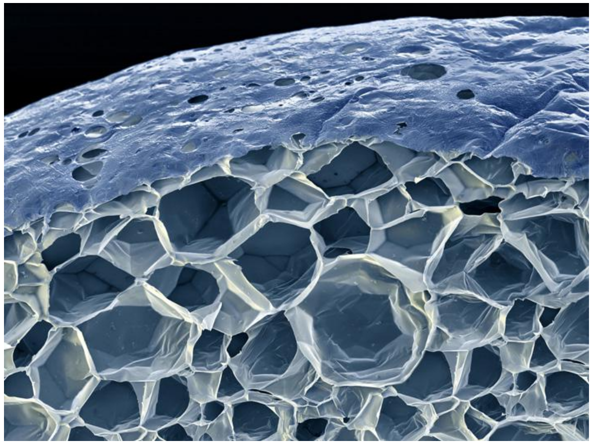
[a] To convert MPa to psi, multiply by 145.
[b] To convert GPa to psi, multiply by 1.45 x 10^5.
[c] To convert J m^{-1} to lb_f in.$^{-1}$, divide by 53.38.

Other styrenic resins

- HIPS
- SAN
- SMA - Styrene Maleic Anhydride
- ~ EPS (expandable polystyrene)

4 to 7% blowing agent, usually pentanes or butane





IAPD THERMOPLASTICS RECTANGLE

IAPD

AMORPHOUS PLASTICS

SEMICRYSTALLINE PLA

IMIDE MATERIALS

Key Characteristics:	Materials:
Very High Cost Per Pound Excellent Properties Above 400¡F Excellent Electrical Properties Excellent Dimensional Stability Low Coefficient of Friction	PI PAI PBI

AMORPHOUS HIGH PERFORMANCE PLASTICS

Key Characteristics	Materials:
High Cost High Temperature High Strength & Good Stiffness Good Chemical Resistance Transparency Hot Water & Steam Resistance	Polysulfone Polyetherimide Polyethersulfone Polyarylsulfone

SEMI-CRYSTALLINE HIGH PERFORMACE PLASTICS

Key Characteristics:	Materials:
High Cost High Temperature High Strength Good Electrical Properties Outstanding Chemical Resistance Low Coefficient of Friction Good Toughness	PVDF PTFE ECTFE FEP PFA PPS PEEK

AMORPHOUS ENGINEERING PLASTICS

Key Characteristics:	Materials:
Moderate Cost Moderate Temperature Resistance Moderate Strength Good Impact Resistance Translucency Good Dimensional Stability Good Optical Qualities	Polycarbonate Modified PPO Modified PPE Thermoplastic Urethane

SEMI-CRYSTALLINE ENGINEERING PLASTICS

Key Characteristics:	Materials:
Moderate Cost Moderate Temperature Resistance Moderate Strength Good Chemical Resistance Good Bearing and Wear Properties Low Coefficient of Friction Difficult to Bond	Nylon Acetal PET PBT UHMW-PE

AMORPHOUS COMMODITY PLASTICS

Key Characteristics:	Materials:
Low Cost Low Temperature Resistance] Low Strength Good Dimensional Stability Bond Well Typically Transparent	Acrylic Polystyrene ABS PVC PETG CAB

SEMI-CRYSTALLINE COMMODITY PLASTICS

Key Characteristics:	Materials:
Low Cost Low Temperature Resistance Low Strength Excellent Chemical Resistance Low Coefficient of Friction Near Zero Moisture Absorption Very Good Electrical Properties Good Toughness	Polyethylene Polypropylene Polymetnylpentene(TPX)

AMORPHOUS PLASTICS KEY CHARACTERISTICS:

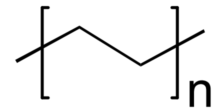
Soften Over a Broad Range Of Temperatures
Easy to Thermoform
Tend to Be Transparent
Bond Well Using Adhesives and Solvents
Prone To Stress Cracking
Poor Fatigue Resistance
Stuctural Applications Only (Not for Bearing & Wear)

SEMI-CRYSTALLINE PLASTICS KEY CHARACTERISTICS:

Sharp Melting Point
Difficult to Thermoform
Tend to Be Opaque
Difficult To Bond Using Adhesives and Solvents
Good Resistance To Stress Cracking
Good Fatigue Resistance
Good For Bearing and Wear, As Well As Structural Applications

Polyethylene

$H-[CH_2-CH_2]_n-H$

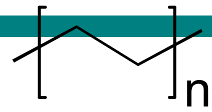


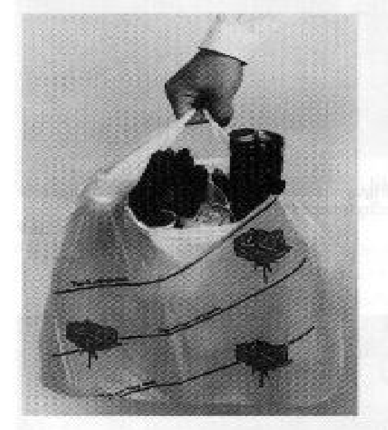
semicrystalline

- Probably most common plastic – glad bags and packing material, children’s toys – thermoplastic
- Simple formula:
- Not amorphous!
- Glass transition -130° to -80° C
- Melting point 130° C
- Tensile yield (strength) 25 MPa
- Tensile modulus (stiffness) 1 GPa (soft)
- Density 0.95

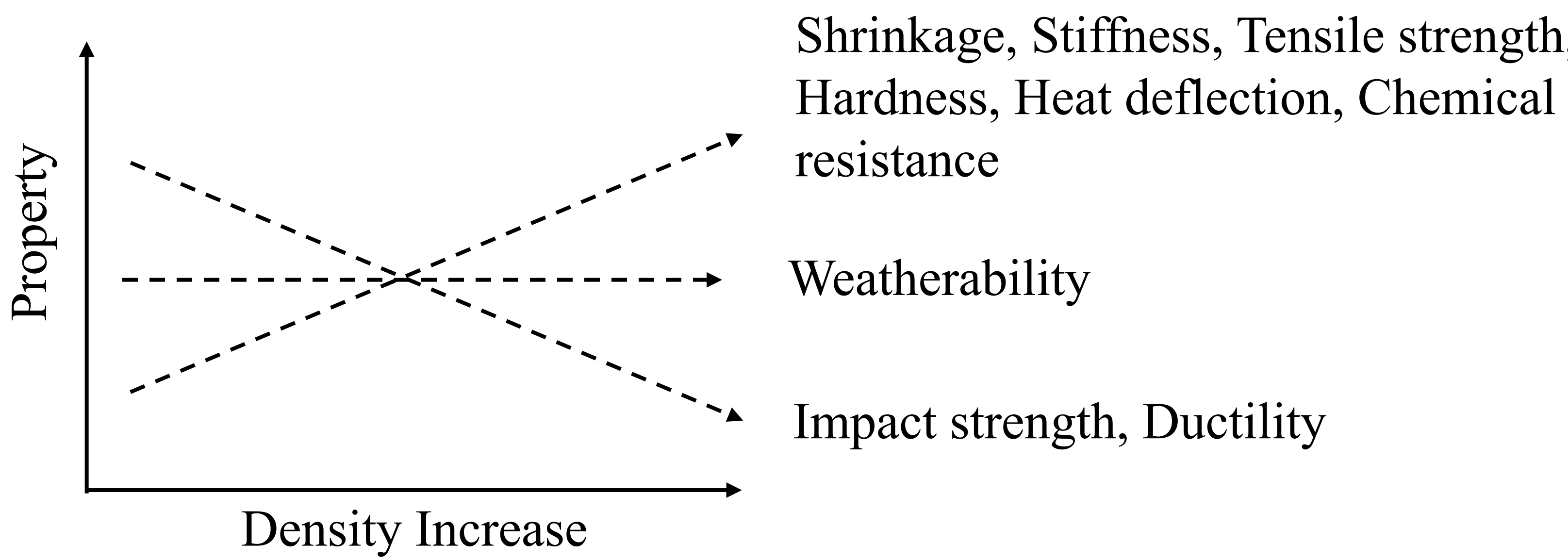
Polyethylene

$H-[CH_2-CH_2]_n-H$



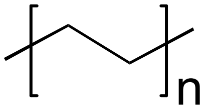


Material	Density (kg/m3)	% Crystallinity
Low density polyethylene (LDPE)	915-929	45-65
Medium density polyethylene (MDPE)	930-939	65-75
High density polyethylene (HDPE)	940-965	75-90

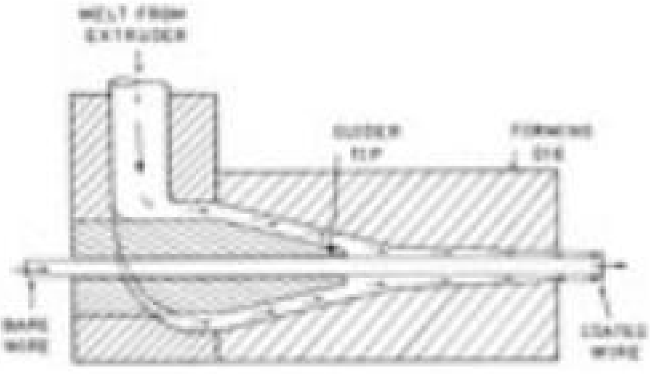


Polyethylene

$H-\!\!\left[CH_2-CH_2\right]_n\!\!-H$



Polymer	Tensile Strength (MPa)	Modulus (MPa)	Elongation (%)
LDPE	8-31	172-283	100-650
HDPE	22-31	1070-1090	10-1200



Wire coating

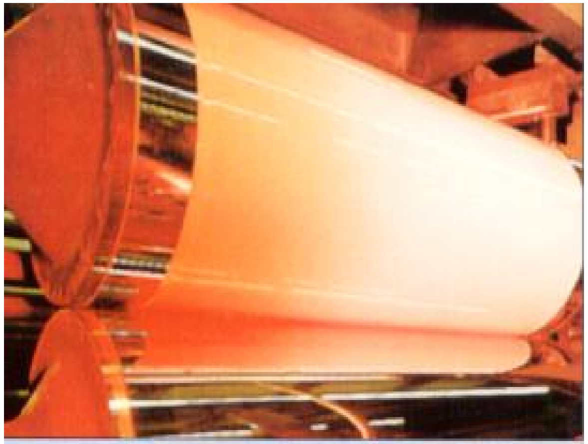
Making Polymer Films

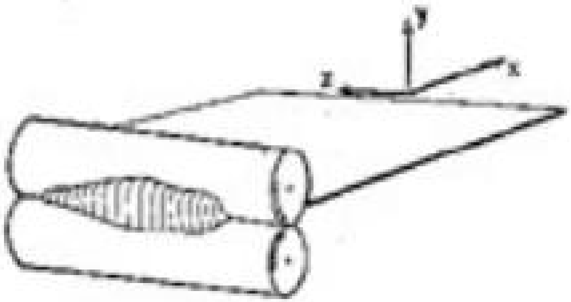
Solvent Free Process (Free standing films)

- Cast Extrusion
- Blow Film Extrusion

Solvent (Coating) Processes

- Spincoating
- Dipcoating
- Liquid spray
- Thermal Spray
- Electrocoating

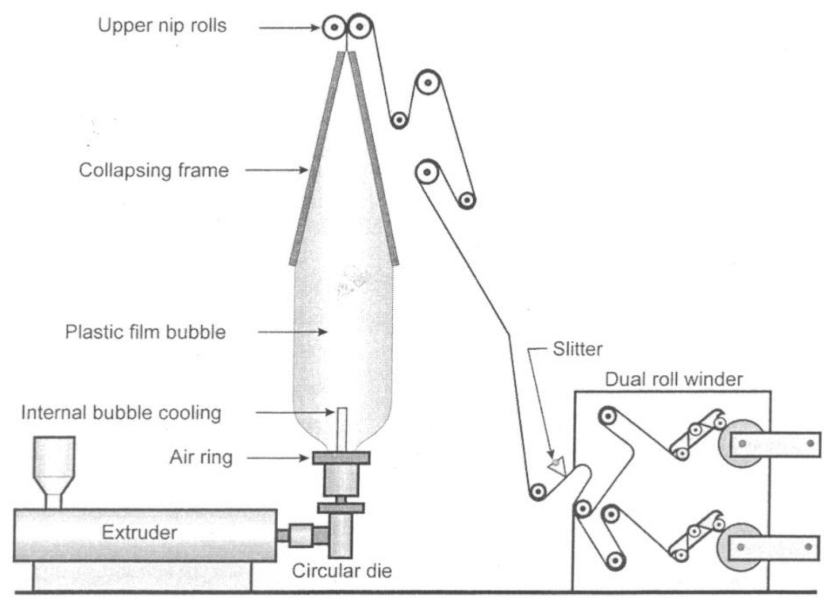




calendaring

Blown Film Extrusion

- Film blowing is the most important method for producing Polyethylene films (about 90% of all PE film produced)
- In film blowing a tubular cross-section is extruded through an annular die (usually a *spiral* die) and is drawn and inflated until the *frost line* is reached. The extruded tubular profile passes through one or two air rings to cool the material.
- Most common materials: LDPE, HDPE, LLDPE



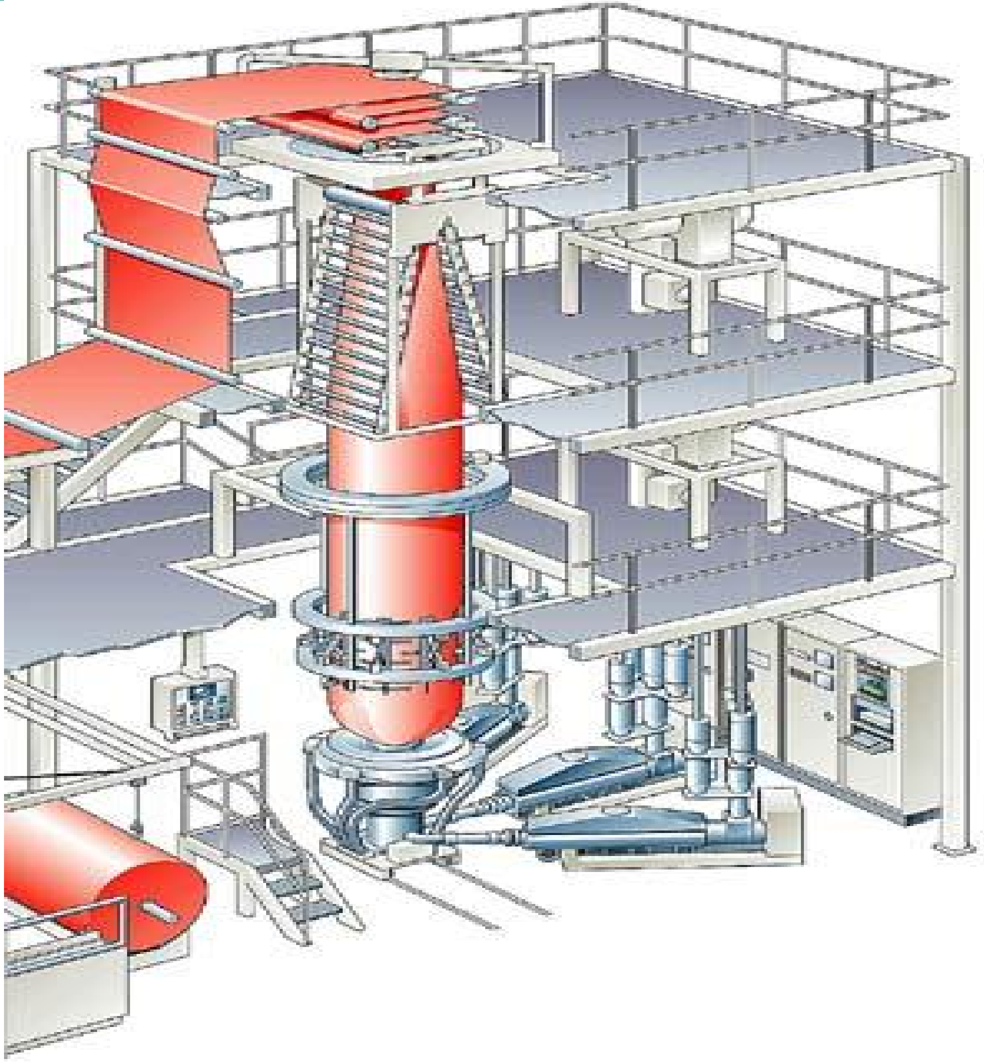
Polyethylene

$T_g = -124\ ^\circ C$ $T_m = 137\ ^\circ C$

Blown film process

http://www.bpf.co.uk/bpfindustry/reifenhauser_gallery.cfm

Blown film process



http://www.56.com/u74/v_MTEzNDMxNTM1.html

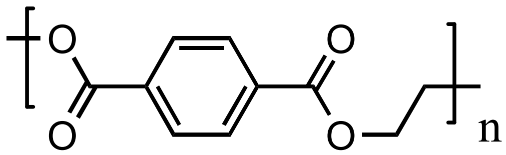
Sheet & film extrusions
Cast vs. blown film

Comparison of films made by the blown and cast processes.

	Blown Film	*Cast Film*
Required melt flow rate	lower	higher
Process temperature	lower	higher
Crystallinity	more (slower cooling)	less (rapid cooling)
Optics	hazy, less gloss	better, glossy
Gauge variation	± 5 to 10%	± 2%
Output	lower	higher
Web control	size adjustable	neck-in, more rework
Flatness	less	better

Blown film is less expensive and more widely used

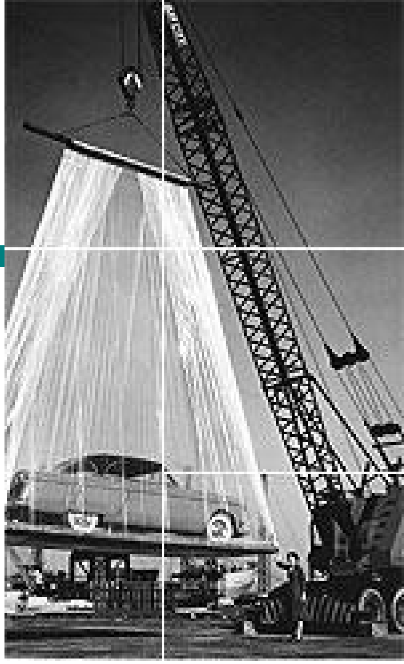
Polyester films



semicrystalline

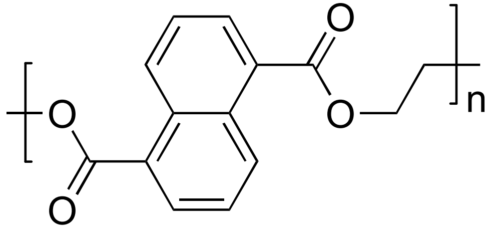
Mylar = PETE Film

Dacron = PETE fiber

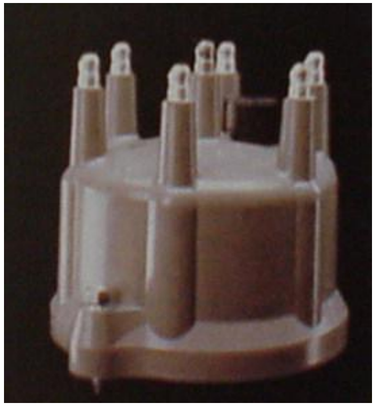


$T_g = 80\ ^\circ C$ $T_m = 260\ ^\circ C$

Tensile Strength: 48-72 MPa, Modulus = 2.7-4.1 Gpa
50-300% elongation



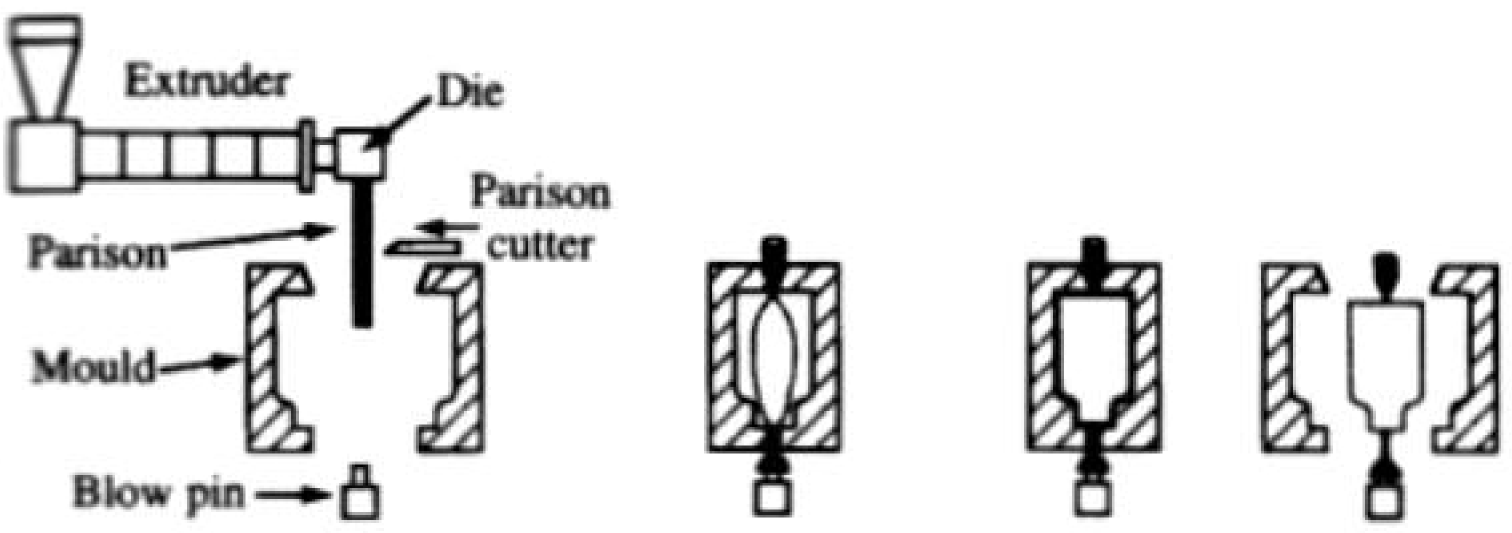
Teonex = PEN Film



$T_g = 120\ ^\circ C$ $T_m = 262\ ^\circ C$

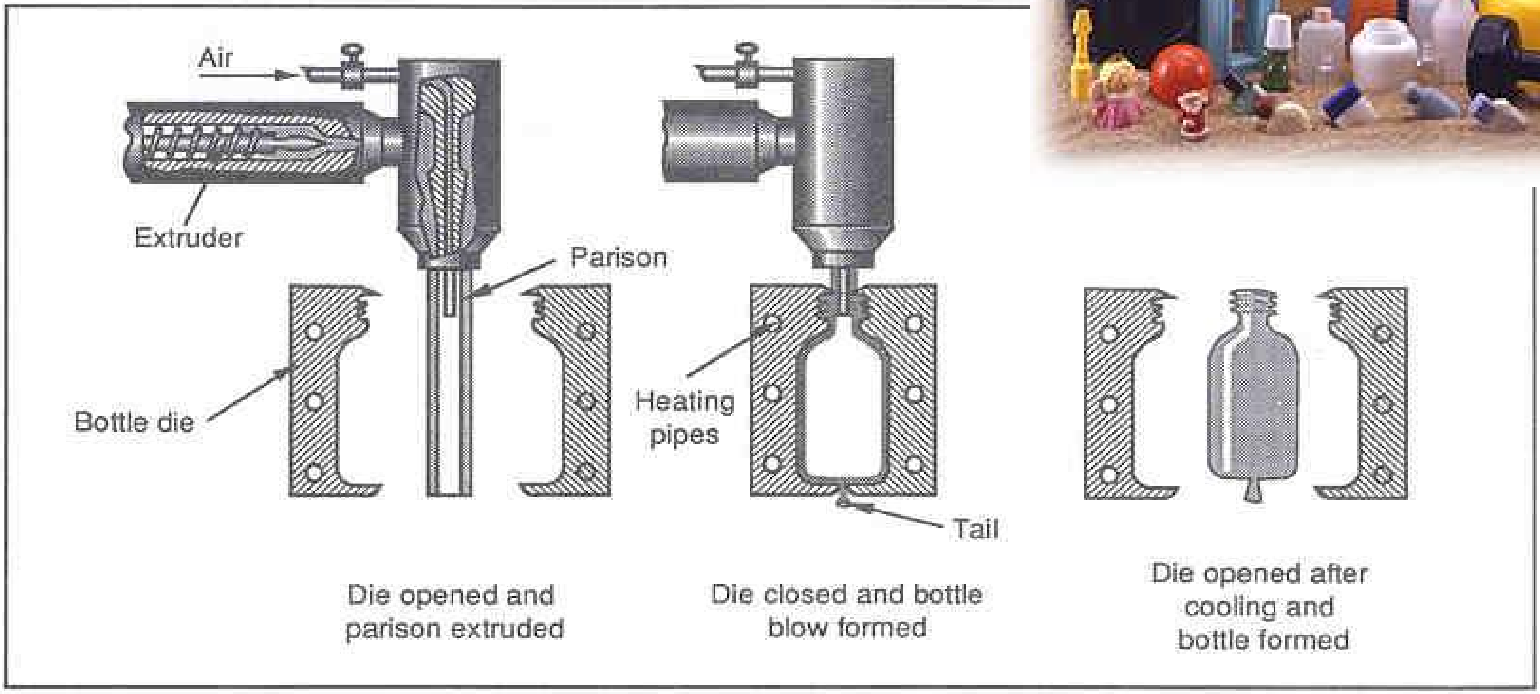
Extrusion Blow molding

In extrusion blow molding, a ***parison*** (or tubular profile) is extruded and inflated into a cavity with a specified geometry. The blown article is held inside the cavity until it is sufficiently cool.



(a) Parison descends (b) Inflation (c) Cooling (d) Ejection

Blow Molding, Bottles



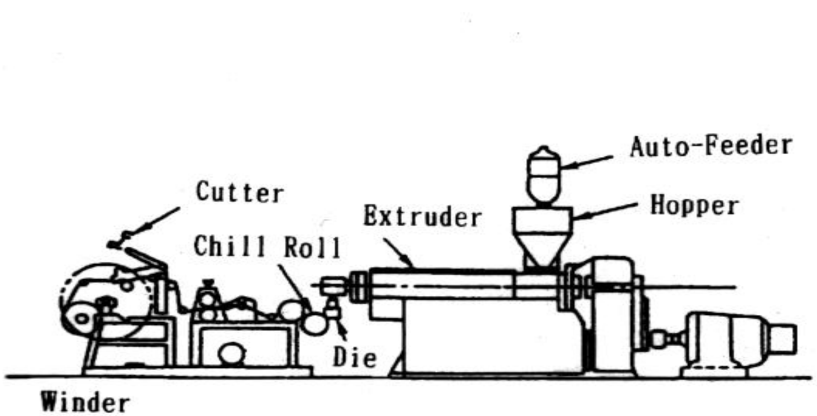
- Hollow plastic parts with relatively thin walls.
- Typical parts: Bottles, Bumpers, Bags, Ducting.

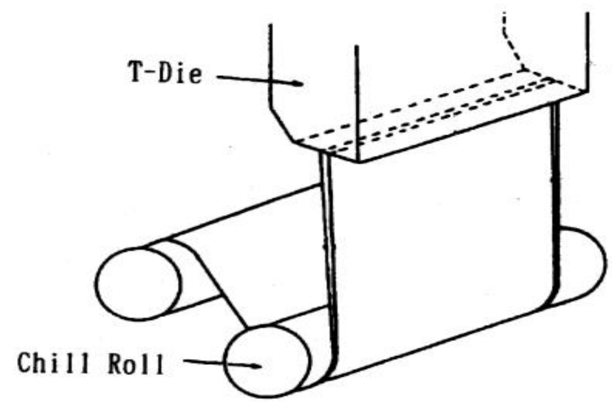
http://my.tv.sohu.com/us/16360744/8821339.shtml

Blow Molding

Cast Film Extrusion

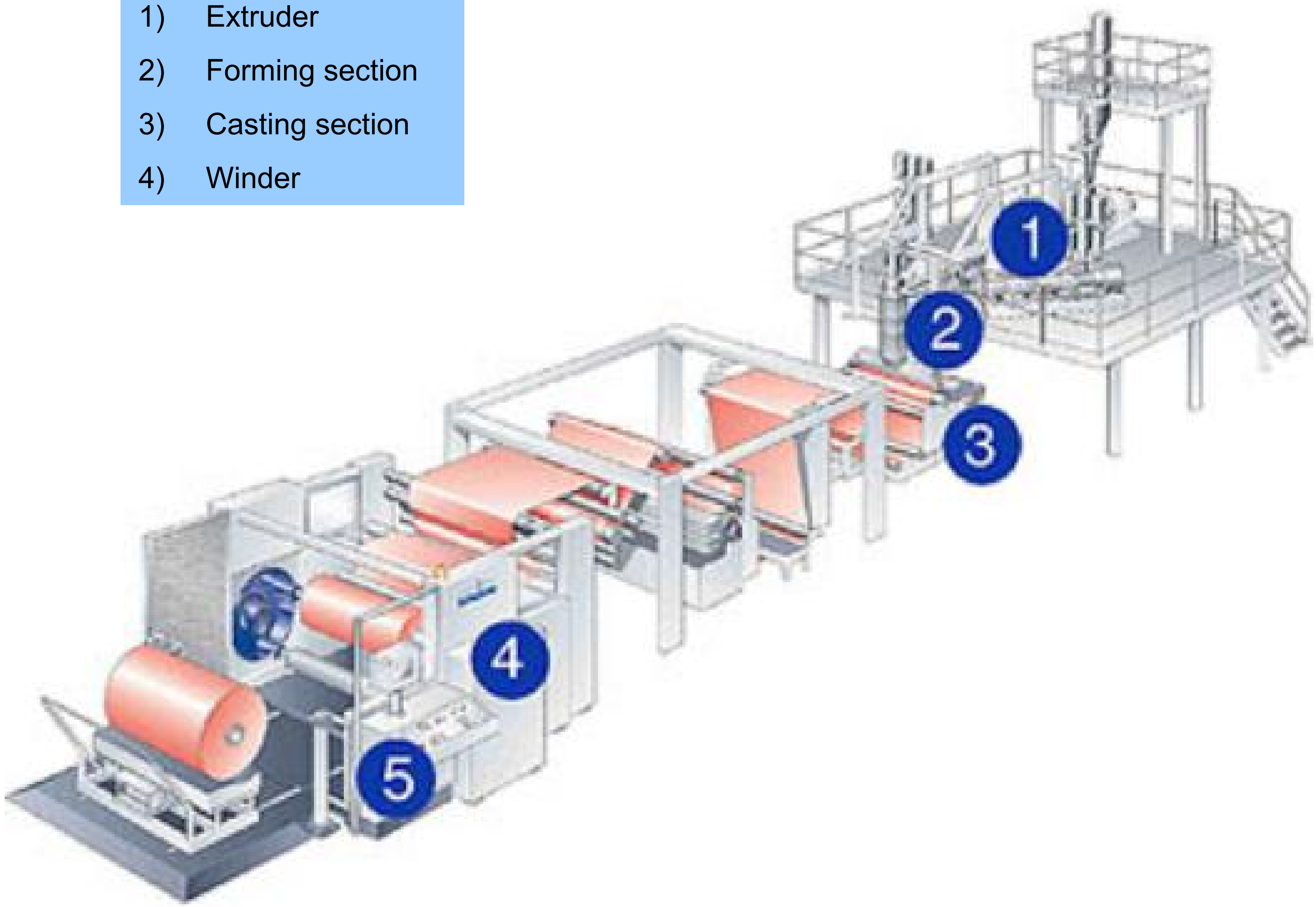
- In a cast film extrusion process, a thin film is extruded through a slit onto a chilled, highly polished turning roll, where it is quenched from one side. The speed of the roller controls the draw ratio and final film thickness. The film is then sent to a second roller for cooling on the other side. Finally it passes through a system of rollers and is wound onto a roll.
- Thicker polymer sheets can be manufactured similarly. A sheet is distinguished from a film by its thickness; by definition a sheet has a thickness exceeding 250 μm. Otherwise, it is called a film.





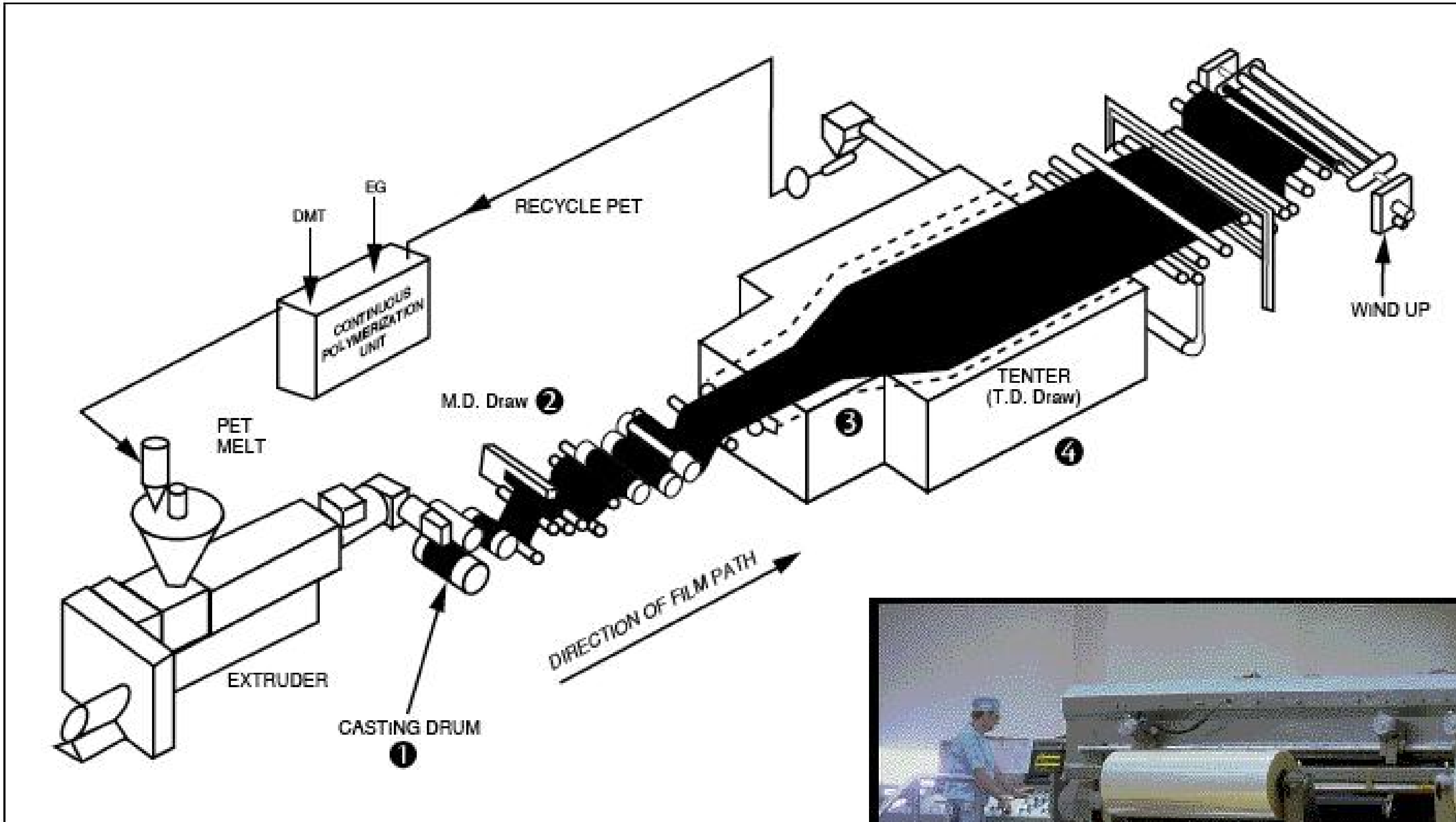
3 layer cast film line

1) Extruder
2) Forming section
3) Casting section
4) Winder



http://www.reifenhauser.com/en/folienanlagen/p0010_flach_mod.asp#5

Figure 1. A Typical Manufacturing Process for Oriented Polyester Film



One side is normally microscopically smooth, while the other side contains microscopic asperities which promote adhesion of coatings and printing media.

Polyester films

Mylar® polyester films have a unique combination of physical, chemical, thermal, and optical properties:

- Strong, tough, brilliant, and clear.
- Ease of converting:

  laminating, extrusion coating, embossing, metallizing,
  printing, punching, corrugation, dyeing,
  stamping or forming.
  Ease of handling on high-speed equipment.
- Retain mechanical properties:

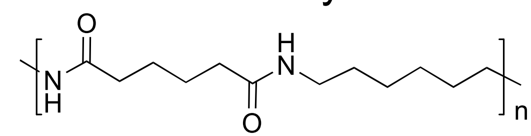
  stiffness, strength, toughness, dimensional
  stability, and optical clarity, over an exceedingly
  wide range of temperatures.
- Excellent temperature resistance.
- Readily combined with other materials.
- Strong tear-initiation and puncture resistance.
- Excellent oil, grease, or moisture barrier resistance.
- Excellent chemical resistance.

Nylon 6,6

- –Excellent wear resistance & slick surface
- –Poor dimensional stability & high cost
- –Gear, engine fan

Strengths:
Good Toughness & Strength
Good Chemical resistance

semicrystalline



Nylon 6,6

mp 265 °C
tg 50 °C

Limitations:

Strong acidic environments

Areas where moisture absorption is of concern

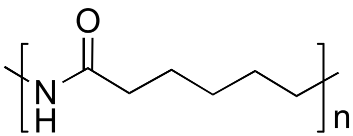
-20% strength with humid environment

Areas experiencing high operating temperatures

Nylon 6: Tensile yield 76 MPa; Tensile modulus 1.4 GPa, elongation 250%
Nylon 6,6: Tensile yield 80 MPa; Tensile modulus 2 GPa, elongation 200%

Interchangable for most applications

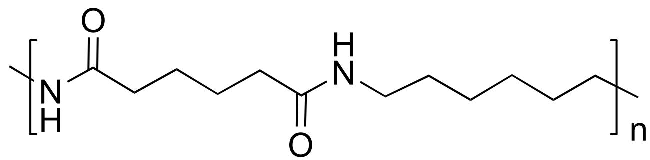
Nylon & Clay nanocomposites



Nylon-6

mp 220 °C

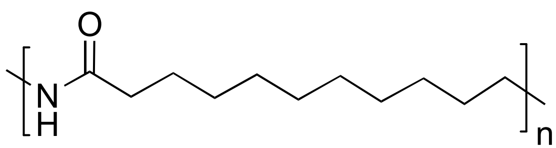
tg 53 °C



Nylon 6,6

mp 265 °C

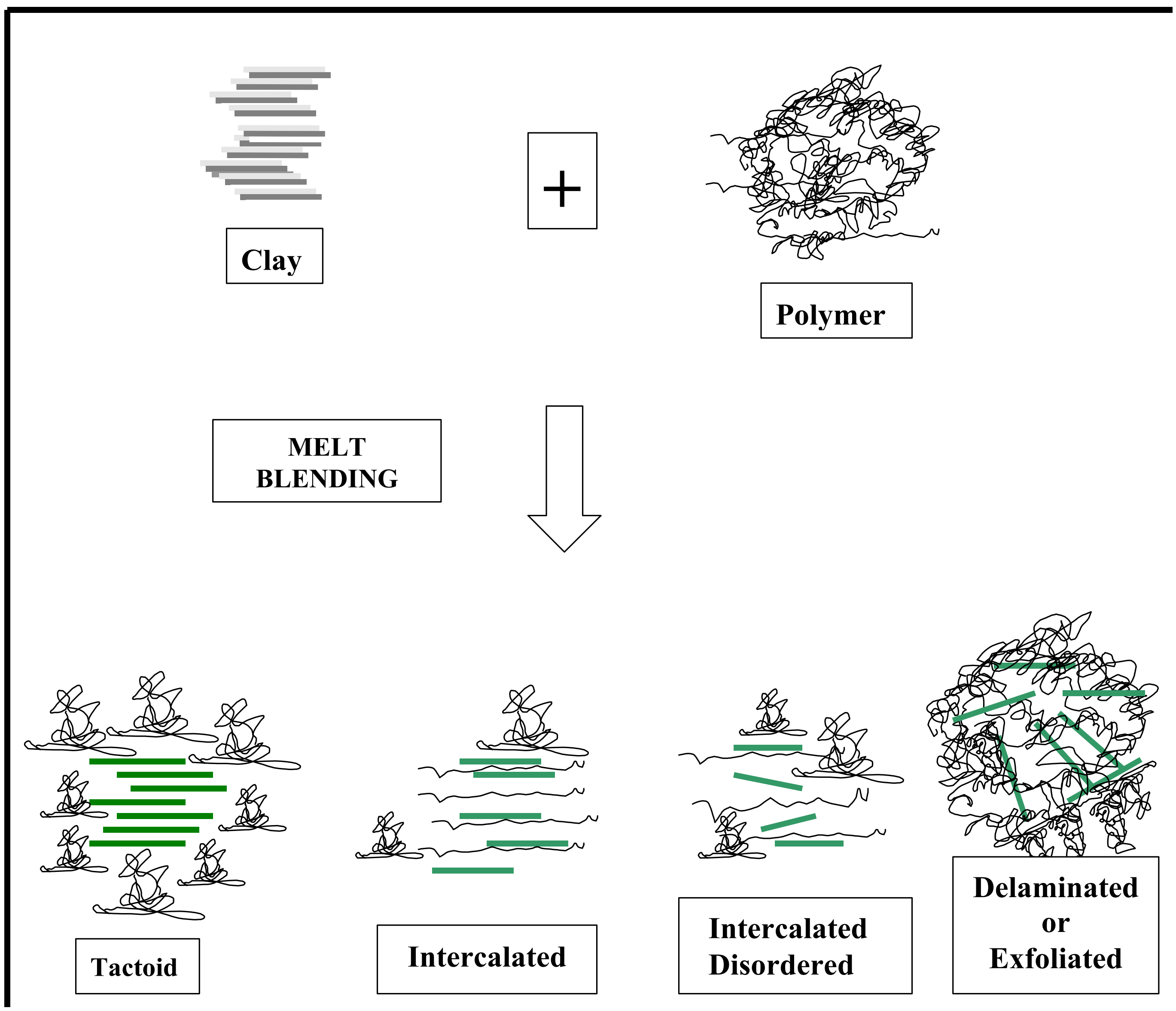
tg 50 °C



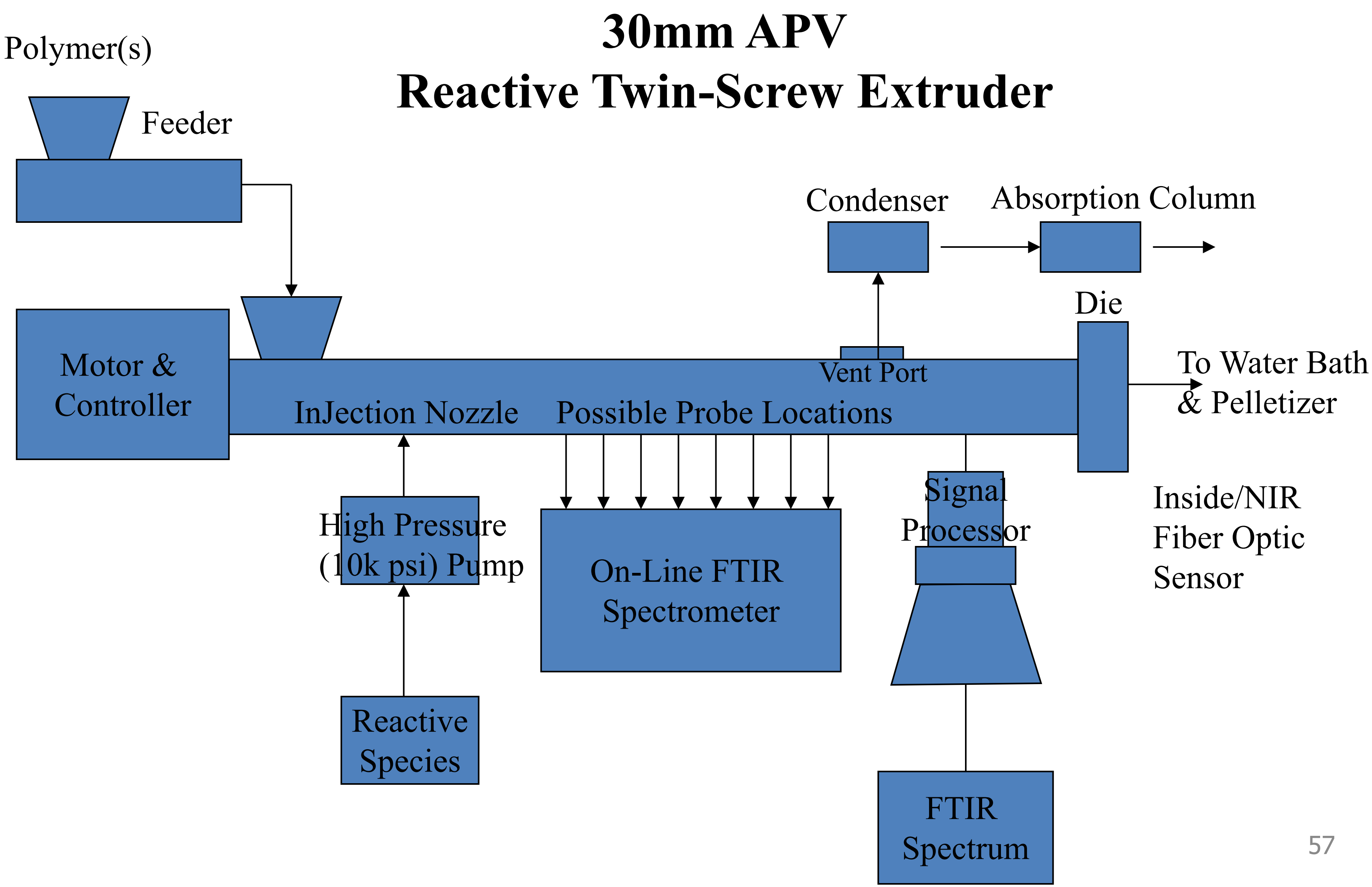
Nylon-11

mp 194 °C

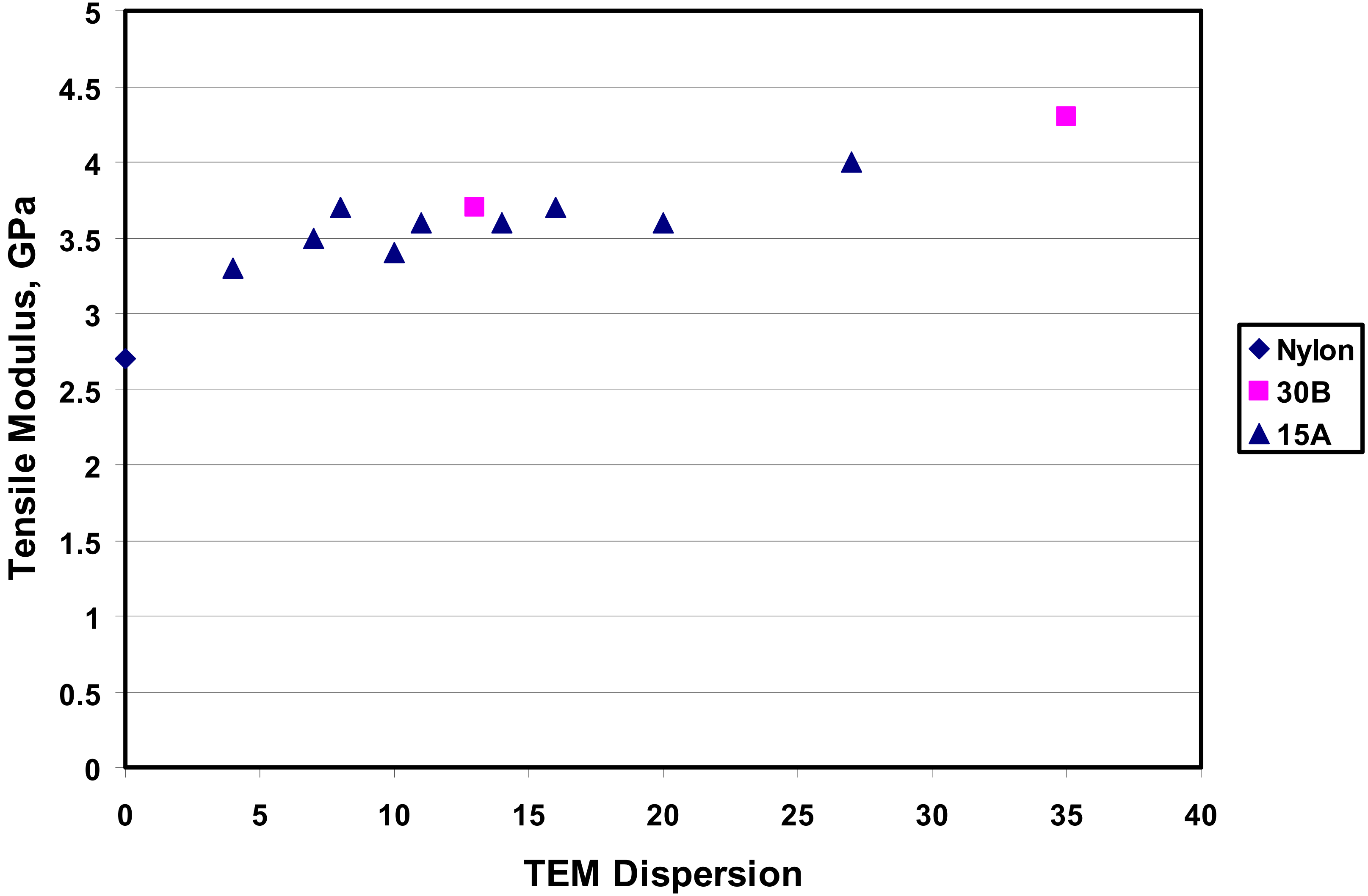
tg 44 °C


Clay
+
Polymer
MELT BLENDING
Tactoid
Intercalated
Intercalated Disordered
Delaminated or Exfoliated

Reactive Extruder



Modulus vs Dispersion



Better barrier properties. Why?

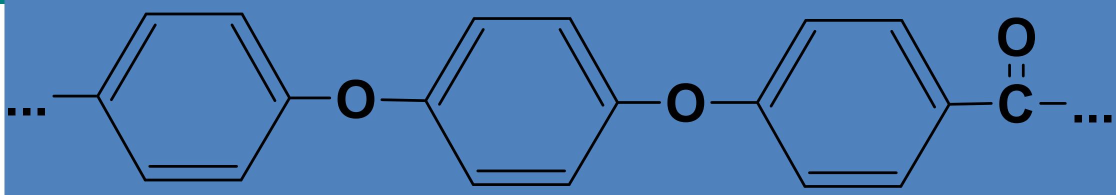
Advanced Thermoplastics

- All highly aromatic
- Most common types:
  - Thermoplastic polyimides (TPI, PAI, PEI, PBI)
  - Polyetheretherketone (PEEK, and related)
  - Sulfone and related (PSU, PAS, PES, PPS)
  - Liquid crystal polymers (LCP)
  - Fluoropolymers (PTFE, PVDF)

High Performance Thermoplastics

Tensile Strength: 65 MPa

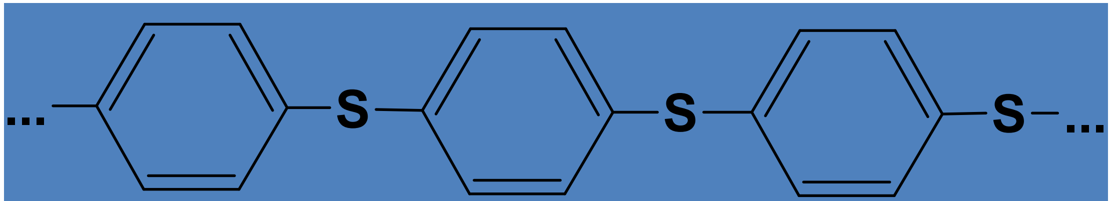
Modulus: 3 GPa



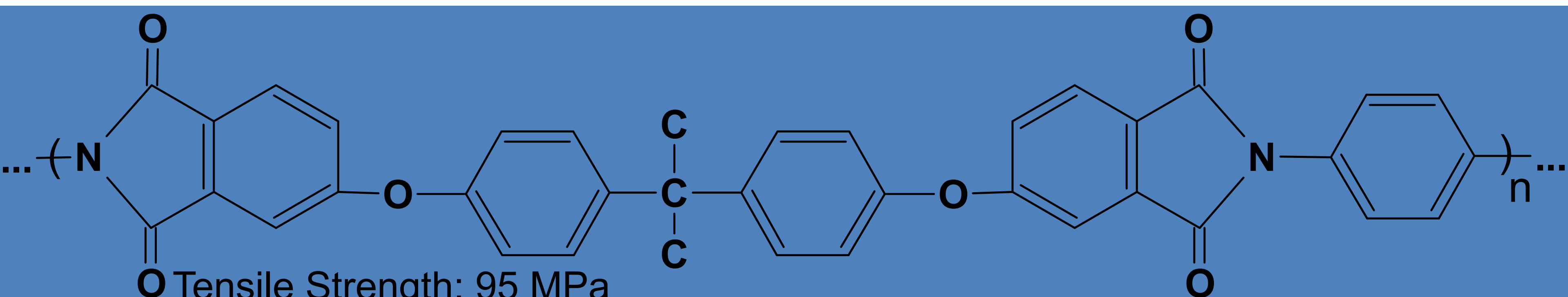
Polyether ether ketone (PEEK)

Tensile Strength: 75 MPa

Modulus: 3.4 GPa



Polyphenylene sulfide (PPS)



Tensile Strength: 95 MPa

Modulus: 25 GPa

Polyetherimide (PEI)

Teflon

- PTFE – Polytetrafluoroethylene – aka Teflon
  long name, simple structure:

semicrystalline

$$-\left[CF_2-CF_2\right]_n-$$

- Exceptional resistance to
  solvents, great lubricant, nothing sticks to it!
- The fluorine-carbon bonds are very strong, fluorines protect carbon backbone.
- High melting point 330° C
- High electrical breakdown – artificial muscle.
- Technically a thermoplastic, but hard to process.
- Opaque due to crystallinity

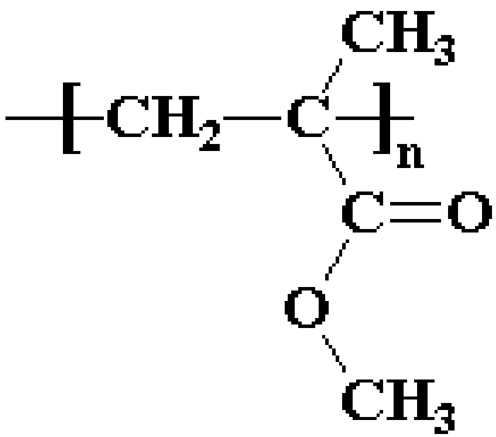
Tensile Strength: 30 MPa

Modulus: 410 MPa

350% elongation

“Acrylic” or Polymethyl methacrylate

- Most common optical plastic - refractive index very close to glass (1.5), aka Plexiglas, Lucite
- Amorphous
- Full name polymethyl methacrylate (PMMA).
- Also an important fiber, paint.
- Glass transition 110° C
- Melting point 130° C
- Tensile yield 50 MPa; Tensile modulus 3 GPa
- 2% elongation



Amorphous

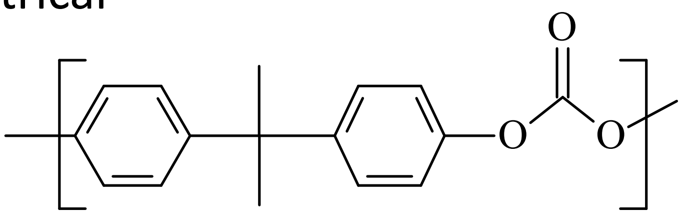
Major applications of PMMA include:

- ➢Automotive industry - rear lamps, light fixtures
- ➢Acrylic sheet - bathtubs
- ➢Glazing - signs
- ➢Composites - sinks, basins and bathroom fixtures.

Polycarbonates

Amorphous

- Commercial polycarbonates are linear, symmetrical
- polymers that, due to extreme chain
- stiffness, do not crystallize easily from
- the melt and hence are amorphous.



Lexan

- Other polymers are as rigid, others are as transparent, but the bis-phenol A polycarbonate shown above provides these qualities at relatively low cost.

- Also affords good
- unnotched impact
- resistance and electrical
- insulating properties.

Consumption (tonnes)	506 000
Application breakdown (%)	
Glazing	28.7
Electric/electronic	21.8
Optical media	17.8
Lighting	8.7
Appliances	5.2
Transportation	4.3
Packaging	3.6
Recreation	2.6
Medical	2.6
Other	4.7

Tensile yield 56 MPa; Tensile modulus 2.3 GPa
>100% elongation

Polycarbonates

- Excellent clarity
- Excellent toughness
- Good heat resistance
- Excellent electrical properties
- Intrinsic flame-retardancy
- Excellent strength

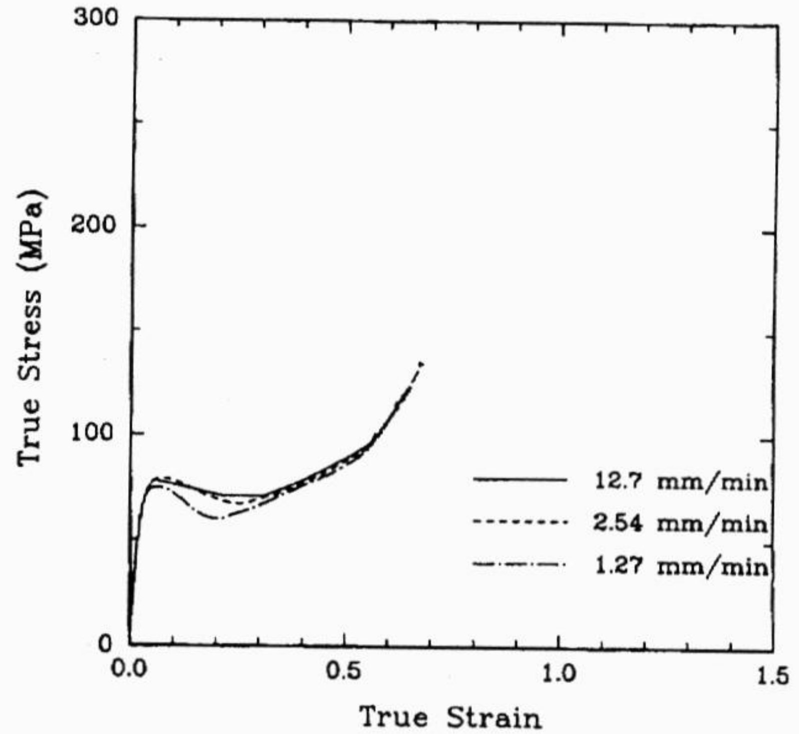


Fig. 6. Local true stress vs. true strain response from the tensile tests of Fig. 4 at the displacement rates of 1.27, 2.54, and 12.7 mm/min.

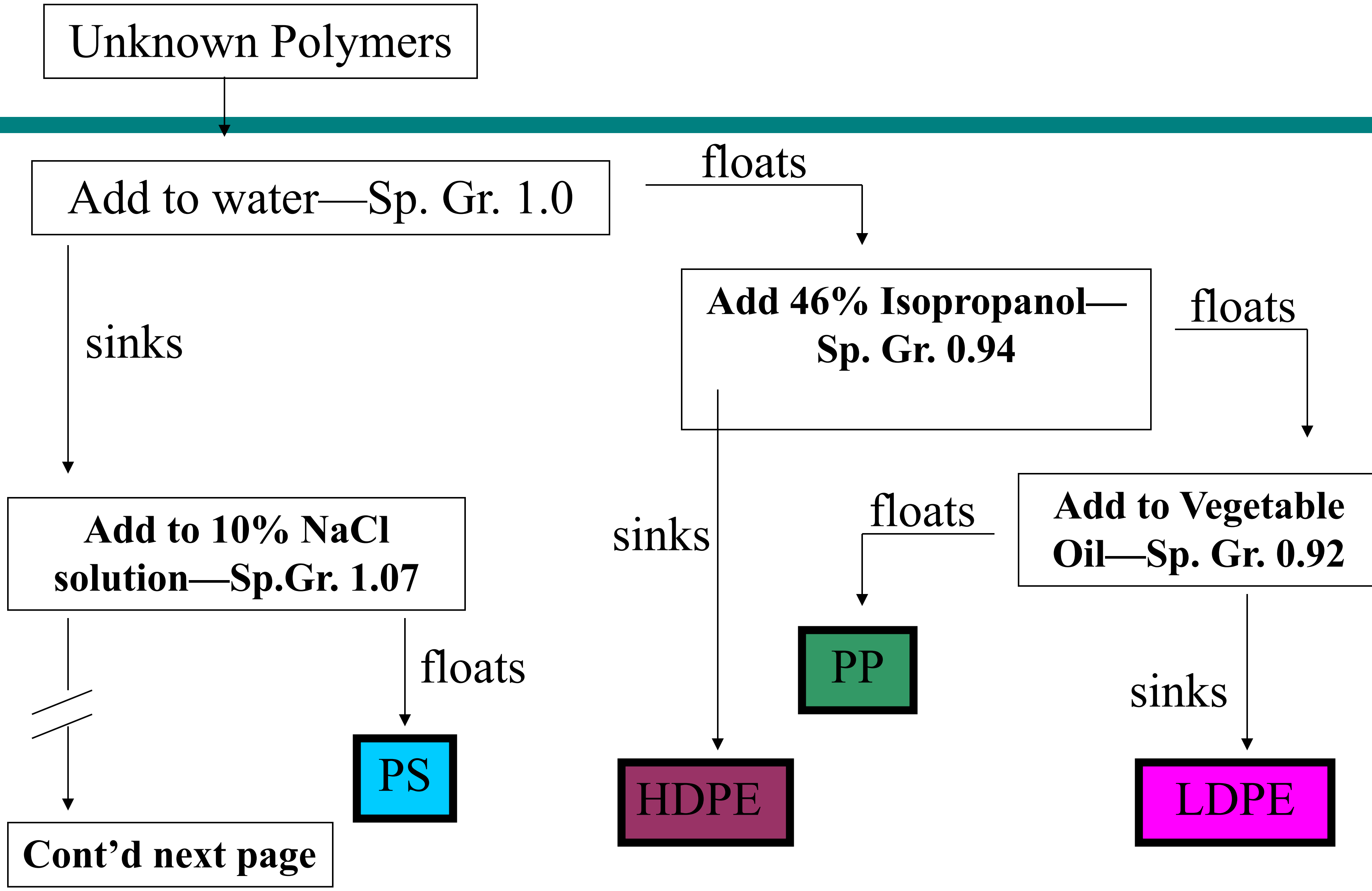
Hot water = gradual embrittlement
Crazed surface with exposure to organic solvents

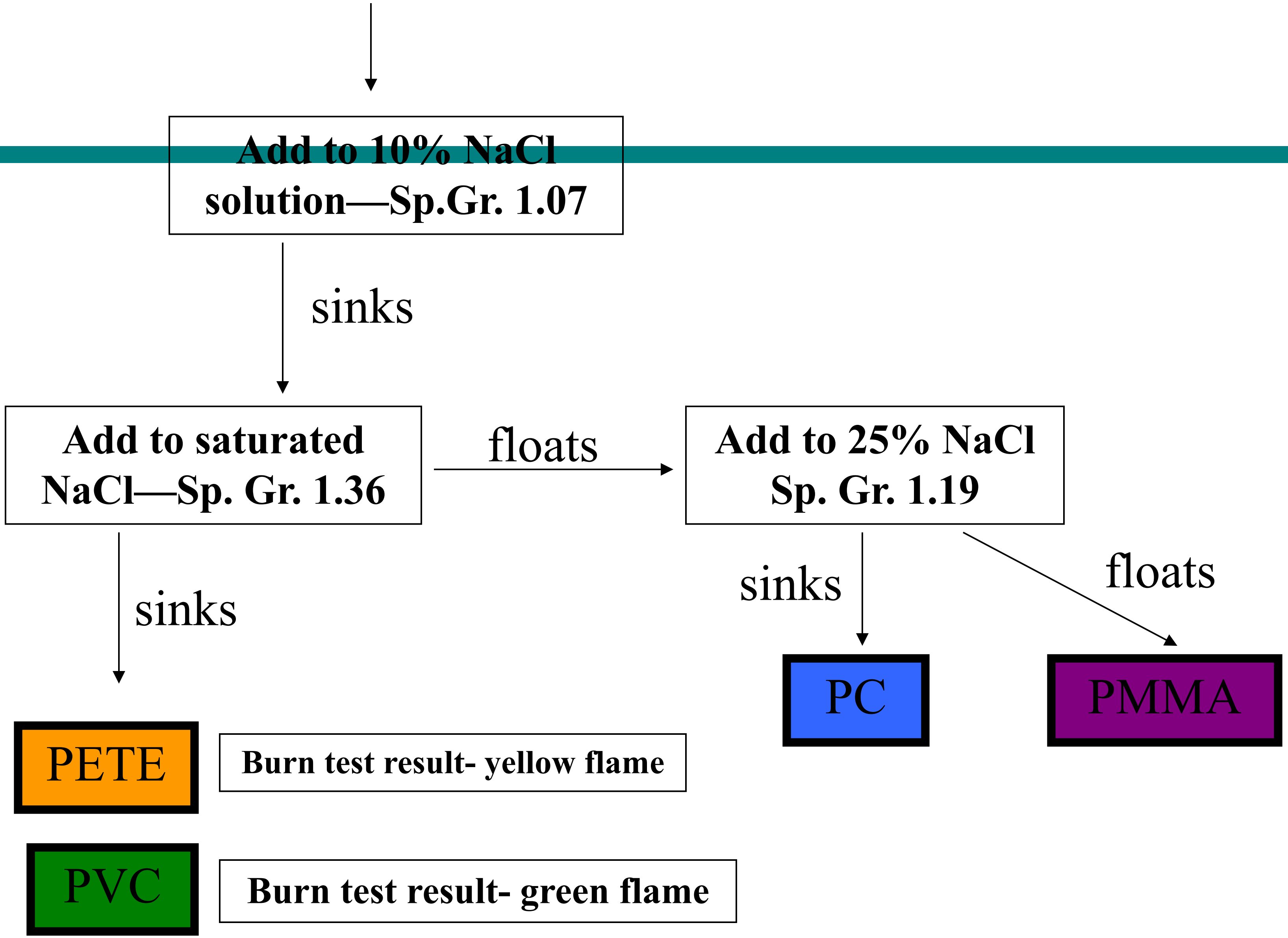
Common Shaping Processes for Thermoplastics

	Compression Molding	Transfer Molding	Injection Molding	Extrusion	Rotational Molding	Blow Molding	Thermoforming	Reaction Injection Molding	Casting	Forging	Foam Molding	Reinforced Plastic Molding	Vacuum Molding	Pultrusion	Calendering
Acetal			●	●	●	●	●				●	●			●
ABS			●	●	●	●	●			●					●
Acrylic	●		●	●		●	●		●						●
Cellulose acetate	●		●	●			●								●
Nylon			●	●	●	●		●	●	●		●			●
Polyimide			●												
Polycarbonate			●	●		●	●								●
Polyethylene			●	●	●	●	●			●	●	●			●
Polypropylene	●		●	●	●	●	●		●	●		●			●
Polystyrene			●	●	●	●	●				●	●			●
Polysulfone			●			●	●					●		●	
Polyurethane			●	●	●						●	●	●		●
PVC	●	●	●	●	●	●	●				●	●	●		●
Polyvinyl acetate	●	●	●	●	●	●	●				●				●
Tetrafluoroethylene	●	●	●	●											●

TABLE 4.1. Mechanical Properties of Common Homopolymers[a]

Polymer \ Property	Tensile Properties at Break: *Strength*[b] *(Mpa)*	Tensile Properties at Break: *Modulus*[b] *(Mpa)*	Tensile Properties at Break: *Elongation (%)*	*Compressive Strength*[b] *(Mpa)*	*Flexural Strength*[b] *(Mpa)*	*Impact Strength*[c] *(N/cm)*
Polyethylene, low density	8.3-31	172-283	100-650	-	-	No break
Polyethylene, high density	22-31	1070-1090	10-1200	20-25	-	0.23-2.3
Polypropylene	31-41	1170-1720	100-600	38-55	41-55	0.23-0.57
Poly(vinyl chloride)	41-52	2410-4140	40-80	55-90	69-110	0.23-1.3
Polystyrene	36-52	2280-3280	1.2-2.5	83-90	69-101	0.20-0.26
Poly(methyl methacrylate)	48-76	2240-3240	2-10	72-124	72-131	0.17-0.34
Polytetra-fluoroethylene	14-34	400-552	200-400	12	-	1.7
Nylon 66	76-83	-	60-300	103	42-117	0.46-1.2
Poly(ethylene terephthalate)	48-72	2760-4140	50-300	76-103	96-124	0.14-0.37
Polycarbonate	66	2380	110	86	93	9.1


Unknown Polymers
Add to water—Sp. Gr. 1.0
floats
sinks
Add 46% Isopropanol—Sp. Gr. 0.94
floats
sinks
Add to 10% NaCl solution—Sp.Gr. 1.07
floats
Add to Vegetable Oil—Sp. Gr. 0.92
floats
sinks
PP
PS
HDPE
LDPE
Cont'd next page


Add to 10% NaCl solution—Sp.Gr. 1.07
sinks
Add to saturated NaCl—Sp. Gr. 1.36
floats
Add to 25% NaCl Sp. Gr. 1.19
sinks
sinks
floats
PC
PMMA
PETE
Burn test result- yellow flame
PVC
Burn test result- green flame